

The Effect of Purchase Intention Interruption on Novelty-Seeking: Evidence from a Large Coffee Chain*

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Abstract

As product innovation has become a required competitive advantage for firms, understanding the driver of consumers' willingness to explore is crucial for new production diffusion. This paper asks how an unexpectedly blocked purchase affects consumers' subsequent novelty-seeking behavior. We study this question using 18 temporary exogenous store closures at a large digital coffee retailer. Because the same closure is more likely to interrupt an intended purchase for some consumers than for others, we use pre-closure behavior to predict each consumer's purchase intention during the closure and estimate a triple-differences design comparing high- and low-predicted-intention consumers. We measure novelty-seeking as the share of distinct products purchased in a period that the consumer has not purchased before. After reopening, the treated-control change in novelty-seeking is 4.15 percentage points lower among consumers with high predicted purchase intention than among those with low predicted intention, equivalent to 8.3% of the pre-period mean. The corresponding triple difference is -4.17 percentage points when exploration is measured using recently introduced products. Consistent with an interrupted purchase goal triggering counterfactual thinking and increasing the salience of familiar choices, the decline in exploration is also stronger among consumers with lower baseline novelty-seeking. More broadly, these findings show that supply interruptions can reduce consumers' receptivity to innovation even after access is restored, implying that firms can improve product-launch targeting, recommendations, and promotional incentives by accounting for whether consumers' likely purchases were recently interrupted.

Keywords: Novelty Seeking, Purchase Interruption, Triple-Difference, Counterfactual Thinking, Goal Persistence

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1 Introduction

Ed Sander, a Dutch technology analyst who frequently travels to China, recalled his first impression of a Mascarpone Cheese Latte offered by Luckin Coffee: “It had like the slightly salty, but cheesy kind of flavor to it, on paper it tasted quite awful. But it tasted amazing.” Sander later posted on X that he was going “cold turkey from the Mascarpone Cheese Latte I’ve been having daily when I’m in China.” One year later, he was still mourning the drink (Emont, 2025). Sander is not alone in a world with continual experimentation and product innovation. Luckin, the market leader, opened 6,000 new stores and introduced 119 new products in 2024 alone, although some of the flavor combinations being developed are described by a Colombia coffee consultant as “heresy” by the standards of his native Colombia (Emont, 2025). To catch up the competitive advantage, Starbucks introduced 78 new beverages in China, 2024 (Wang, 2024). In fast food, Taco Bell tests hundreds of ideas each year and uses limited-time offers to keep menus and apps populated with new items that can bring customers back; products that take years to develop may remain on the menu for just four to six weeks (Haddon, 2025).

Product innovation, however, is not self-executing. With a frequent menu shift and a large assortment, firms need consumers to depart from familiar choices and try something new. At the same time, frequent product changes create a common problem to consumers of losing access to their routine of familiar choices, a scenario that could also be caused by inventory shortage or store closure. The supply interruption may not only affect whether consumers come back again, but also their new product adoption behavior in the future, which has already been proved to vary with marketing strategy, market conditions, and consumer characteristics (Steenkamp and Gielens, 2003). Understanding the impact of supply interruption on consumers’ novelty-seeking behavior and its underlying mechanism is thus managerially relevant and important for firms to better conduct product launches (Talay et al., 2024; Ward, 2025) and design related marketing strategies such as new product notification and coupon issuing.

In this paper, we ask: What happens to consumers’ willingness to explore when a purchase they likely intended to make is unexpectedly blocked? This question speaks directly to the tension described above: when firms continuously introduce new products while consumers simultaneously develop routines around familiar choices, an interruption to an intended purchase may alter consumers’ subsequent receptivity to novelty. Answering this question, however, poses two main empirical challenges. First, purchase intention is inherently unobserved when no purchase occurs: a consumer who had no intention to purchase and a con-

sumer whose intended purchase was blocked both leave the same observational outcome of no transaction, making it difficult to distinguish these two. Second, identifying the subsequent impact on novelty-seeking requires longitudinal observational data that follow consumers before and after such an interruption, while market conditions evolve simultaneously over time. Firms expand their store networks, product assortments change, and new products are continuously introduced, all of which can independently affect consumers' opportunities and incentives to explore. Addressing these challenges therefore requires a way to identify consumers who were likely to purchase, together with an external disruption that prevents some of these consumers from completing their intended purchases.

We study this question in a setting of a large Chinese quick-service coffee chain whose consumers order through a mobile application for pickup or delivery. The chain offers a broad and frequently changing menu, creating repeated opportunities to choose between familiar and unfamiliar products. Our data contain 7.3 million orders from 779,744 consumers at 260 stores between June 1, 2020 and December 31, 2021. The empirical exogenous variation comes from 18 temporary store closures lasting 10 to 29 days during COVID-19-related lockdowns. Similar shocks have also been leveraged by prior literature ([Seenivasan et al., 2016](#); [Levine and Seiler, 2023](#); [Ye and Shankar, 2025](#)). A temporary closure can affect subsequent product exploration through several channels. Product assortments and marketing activities may change around reopening, while mobility restrictions may alter consumers' shopping routines and consumption occasions. Reduced purchasing during the closure can also leave more products untried in consumers' purchase histories. Our question is whether the post-reopening response additionally depends on the likelihood that the closure interrupted a purchase the consumer otherwise would have made. The same closure exposes consumers to a supply disruption, but it is more likely to interrupt an intended purchase for consumers with a higher propensity to purchase during the closure window. Using only pre-closure behavior, we predict this propensity and compare the treated-control change in novelty-seeking between consumers with high and low predicted purchase intention in a triple-differences design. This comparison focuses on the incremental response associated with a higher likelihood of purchase interruption. This approach complements recent work that uses hurricane-induced stockouts and temporary store disruptions as exogenous changes in choice opportunities to study persistence in realized brand and store choices ([Levine and Seiler, 2023](#); [Levine and Hristakeva, 2026](#)).

We find that interrupting a likely purchase substantially narrows subsequent product exploration. We measure novelty-seeking as the share of distinct products purchased in a period that the consumer has not

purchased before. Relative to consumers with low predicted purchase intention, consumers whose purchases were more likely to be interrupted exhibit a 4.15-percentage-point lower post-reopening response in novelty-seeking, equivalent to 8.3% of the pre-period mean. The result is nearly identical when we define exploration using recently introduced products: the corresponding reduction in new-product-seeking is 4.17 percentage points. We interpret the pattern through familiar-choice salience: an interrupted purchase leaves a counterfactual thinking of the potential purchase if there were no interruption and a salient representation of the unfinished goal; when consumers get access to the purchase again, that representation can pull choice toward the familiar purchase path and away from exploration. The heterogeneity evidence further supports this interpretation. The reduction in novelty-seeking is strongest among consumers who were less exploratory before the closure, and the estimated gradient is similar with and without adjustment for the amount of baseline purchase history. We also examine two alternative explanations: differential exposure to new-product notifications and differences in the assortments consumers encounter upon returning, and find that neither is supported by extra evidence in data.

Our results offer several managerial implications for product launch and post-disruption marketing strategies. Firms commonly evaluate product launches based on product characteristics, competitive conditions, and market-wide timing, but our findings suggest that consumers' recent purchase experiences can also shape their receptivity to new products. After a store closure, stockout, app outage, delivery suspension, or other temporary access constraint, consumers whose likely purchases were interrupted become less willing to explore unfamiliar products and more inclined toward familiar choices. Firms may therefore want to account for such heterogeneity when deciding whom to target with newly introduced products. When launch timing or targeting is flexible, early promotion of new products may be directed toward consumers whose purchases were less likely to have been interrupted, while familiar or previously purchased products can be recommended to interrupted consumers to better match their immediate preference for familiar choices. When rapid new-product diffusion is strategically important and these consumers cannot be avoided as a target segment, firms may need to provide stronger incentives, such as coupons, price promotions, or other trial inducements, to encourage them to depart from their familiar purchase path. The findings also have implications for personalized recommendation systems: rather than uniformly promoting recently introduced products after operations recover, firms can incorporate whether a consumer's likely purchase was interrupted when deciding between recommending familiar and novel products. More broadly, recovered traffic or transaction volume does not necessarily imply recovered receptivity to innovation. Firms managing dis-

ruptions should therefore consider both the level and the composition of returning demand when designing product launches, recommendations, promotions, and new-product diffusion strategies.

This paper contributes three streams of prior literature. First, we contribute to research on novelty-seeking and exploratory consumer behavior. Novelty-seeking refers to consumers' tendency to approach products or consumption experiences that are new to them and has long been viewed as an important dimension of consumer innovativeness and exploratory behavior (Hirschman, 1980; Raju, 1980; Steenkamp and Baumgartner, 1992). This literature offers several explanations for why consumers seek novelty. Consumers may explore to obtain stimulation and satisfy curiosity (Raju, 1980; Steenkamp and Baumgartner, 1992), to learn about uncertain product quality (Erdem and Keane, 1996), or because they differ in their underlying propensity to adopt and experiment with new products (Hirschman, 1980; Steenkamp and Gielens, 2003). Research on new-product adoption further shows that consumers' willingness to try unfamiliar products depends not only on such individual differences, but also on product perceptions, social context, and firms' marketing activities (Ostlund, 1974; Fisher and Price, 1992; Steenkamp and Gielens, 2003). Closely related research on variety-seeking shows that exploratory choice is also responsive to the immediate choice environment, including prior consumption, contextual stimulation, perceived freedom, time of day, choice architecture, and active goals (Simonson, 1990; Menon and Kahn, 1995; Levav and Zhu, 2009; Gullo et al., 2019; O'Donnell et al., 2023; Rafieian et al., 2024). We extend this literature by showing that a prior purchase interruption can also shape subsequent novelty-seeking. We show that an interruption to a familiar purchase can carry over to subsequent choices and reduce consumers' willingness to discover products they have not previously purchased. Thus, novelty-seeking is shaped not only by characteristics of consumers, products, and the contemporaneous choice environment, but also by whether consumers were able to complete a purchase they previously intended to make.

Second, we contribute to research on temporary disruptions. Store closures affect sales, store switching, retention, channel use, and subsequent activity (Haans and Gijsbrechts, 2010; Seenivasan et al., 2016; Ye and Shankar, 2025); stockouts and service-capacity constraints affect substitution, product choice, and later purchasing (Fitzsimons, 2000; Anderson et al., 2006; Jing and Lewis, 2011; Che et al., 2012; von Wangenheim and Bayón, 2007). Recent work further uses temporary changes in choice opportunities to distinguish state dependence from persistent consumer heterogeneity (Levine and Seiler, 2023; Levine and Hristakeva, 2026). We shift the unit of interpretation from the disruption itself to whether the disruption *binds against an active purchase intention*. That distinction allows the same exogenous supply shock to reveal a behavioral

consequence that average closure effects can miss: a purchase that never occurs can nevertheless shape what consumers choose later. In this sense, the disruption is the empirical lever; the contribution is evidence that intended but unrealized consumption can leave a persistent footprint on product exploration.

Third, we provide the field evidence to research on counterfactual thinking, interrupted decisions, and goal persistence. Counterfactual thinking makes salient how an event could have unfolded differently and can guide subsequent intentions and behavior (Roese, 1994; Epstude and Roese, 2008). Research on consumer goals provides a complementary prediction about what happens when pursuit is interrupted: resuming an interrupted decision can shift processing toward a more goal-directed mode (Liu, 2008); a goal that has been set aside can regain influence over subsequent choice (Carlson et al., 2013); and proximity to completing a focal goal can increase the attractiveness of returning to it (Jhang and Lynch, 2015). Recent work further shows that making progress toward an explicit goal can reduce switching to alternative means by increasing the relative attractiveness of the initial means (Friedman et al., 2025). Our setting brings these ideas to naturally occurring repeated purchases. A temporary closure makes the unrealized purchase unusually easy to imagine and leaves the familiar consumption path unresolved. The main effect and the predicted heterogeneity together are consistent with the resulting familiar-choice salience carrying into subsequent marketplace choices and reducing exploration.

2 Conceptual Motivation and Empirical Predictions

2.1 Behavioral Mechanism: Counterfactual Thinking and Goal Persistence

A purchase interruption differs from simple non-purchase. A consumer who did not intend to purchase leaves no purchase goal unresolved. By contrast, when access is unexpectedly removed from a consumer who was likely to purchase, the event prevents an intended consumption episode from being completed. We propose that this unfinished episode carries into subsequent choice and changes the consumer’s willingness to explore.

Counterfactual thinking and research on goal persistence explain why. Counterfactual thinking makes people represent how an event could have unfolded differently and can guide subsequent intentions and behavior (Epstude and Roese, 2008). Unfulfilled goals can remain cognitively accessible (Masicampo and Baumeister, 2011); goals that have been set aside can regain influence over later choice (Carlson et al., 2013); and proximity to goal completion can increase the attractiveness of returning to the focal task (Jhang

and Lynch, 2015). In consumer settings, responses to product unavailability likewise depend on commitment to the intended option (Fitzsimons, 2000). An interrupted purchase can therefore remain psychologically active after the access constraint itself disappears.

For repeated purchases, the direction of this carryover depends on the consumption path associated with the unfinished goal. When an intended purchase follows an established routine, familiar products provide a readily accessible representation of what the consumer would otherwise have chosen. We propose that interrupting such a purchase makes this familiar path salient through counterfactual thinking and the continued accessibility of the unfinished goal. Once access returns, familiar products can therefore become more salient than unfamiliar alternatives. We refer to this response as *familiar-choice salience*. This account predicts a stronger reduction in exploration among consumers whose prior purchases are more concentrated on familiar products.

The same mechanism generates a heterogeneity prediction. Consumers with lower baseline novelty-seeking concentrate more of their observed consumption on familiar products and established routines. For these consumers, an interruption should more strongly reactivate an established familiar path. Consumers who are already highly exploratory have less concentrated familiar routines, so the same interruption should exert a weaker pull away from novelty. Baseline novelty-seeking therefore provides a natural moderator of the interruption effect.

2.2 A Parsimonious Framework

We formalize this intuition with a reduced-form framework that maps the behavioral mechanism to the empirical design. Let i index consumers and t relative periods around the focal closure. Let Post_t indicate periods after reopening, Treated_t indicate whether the consumer is exposed to the focal store closure, and let $p_i \in [0, 1]$ denote the ex-ante probability that consumer i would have made a purchase during the closure window absent the closure. A larger p_i means that the same closure is more likely to interrupt a purchase the consumer otherwise would have made.

Let Y_{it}^N denote novelty-seeking conditional on making a purchase, and let ϑ_i summarize the consumer's underlying tendency to explore unfamiliar products. We represent expected novelty-seeking as

$$\mathbb{E}[Y_{it}^N \mid \text{Purchase}_{it} = 1] = \mu_{it}(\vartheta_i) - \lambda(\vartheta_i)(\text{Post}_t \times \text{Treated}_t \times p_i), \quad (1)$$

where $\mu_{it}(\vartheta_i)$ collects the components of expected novelty-seeking that are unrelated to the interruption-specific familiar-choice channel. The function $\lambda(\vartheta_i)$ captures the strength of familiar-choice salience generated by a likely purchase interruption. The interaction $\text{Post}_t \times \text{Treated}_i \times p_i$ is zero before the closure and for control consumers, and becomes larger after reopening for treated consumers with greater ex-ante purchase propensity. Equation (1) therefore isolates the behavioral component that should vary with the likelihood that the closure actually interrupted an intended purchase.

The behavioral account imposes two restrictions:

$$\lambda(\vartheta_i) > 0, \quad \lambda'(\vartheta_i) < 0. \quad (2)$$

The first captures the central mechanism: a likely purchase interruption shifts subsequent choice toward familiar products and away from exploration. The second captures the predicted heterogeneity: consumers with a stronger underlying tendency to explore have a weaker familiar-choice response to the interruption. In the empirical analysis, we use consumers' baseline novelty-seeking to proxy for this underlying heterogeneity.

For treated consumers after reopening, the interruption-related component of novelty-seeking becomes more negative as purchase intention rises:

$$D_p(\vartheta_i) \equiv \frac{\partial}{\partial p_i} [-\lambda(\vartheta_i)p_i] = -\lambda(\vartheta_i) < 0. \quad (3)$$

Here $D_p(\vartheta_i)$ is the marginal change in the interruption-related novelty response as the probability of an intended purchase rises. A negative value means that, among otherwise comparable treated consumers, a closure that is more likely to bind against an intended purchase should produce a more negative post-reopening novelty response.

Prediction 1 (Purchase interruption reduces novelty-seeking). *Compared with consumers with lower purchase intention during the closure, consumers with higher purchase intention should exhibit a lower treated-control post-reopening response in novelty-seeking. Thus, the high-minus-low triple-differences effect on novelty-seeking should be negative.*

The framework uses the continuous purchase probability p_i to make the intensity of likely interruption explicit. Our main empirical specification instead classifies consumers into high- and low-predicted-

intention groups. For any $p_H > p_L$, the corresponding finite-difference implication is

$$D(p_H, p_L; \vartheta_i) = -\lambda(\vartheta_i)(p_H - p_L) < 0. \quad (4)$$

This expression is the discrete counterpart of equation (3): moving from a lower to a higher purchase propensity makes the interruption-related response more negative by an amount proportional to $p_H - p_L$. Under the identifying conditions developed in Section 4.3, this finite difference maps to the high-minus-low triple-differences estimand.

The second restriction generates the heterogeneity prediction:

$$\frac{\partial D(p_H, p_L; \vartheta_i)}{\partial \vartheta_i} = -\lambda'(\vartheta_i)(p_H - p_L) > 0. \quad (5)$$

Because $D(p_H, p_L; \vartheta_i) < 0$, a positive derivative means that the interruption effect becomes less negative as the consumer’s underlying tendency toward novelty increases.

Prediction 2 (Baseline novelty attenuates the interruption effect). *The negative interruption-related effect on novelty-seeking should be stronger among consumers with lower baseline novelty-seeking and weaker among consumers who are already more exploratory.*

Together, the two predictions organize the empirical analysis: the first tests whether a likely purchase interruption narrows subsequent exploration, and the second tests whether that narrowing is strongest when consumers’ prior choices are more concentrated on familiar products.

3 Setting and Data

Our setting combines three features that are particularly useful for studying purchase interruption and novelty-seeking: detailed digital purchase histories, a broad and frequently changing product menu, and temporary localized closures that sharply restrict access for a subset of consumers. We use administrative records from a large Chinese quick-service coffee chain whose consumers order primarily through a mobile application for pickup or delivery. The data cover 19 months from June 1, 2020 to December 31, 2021 and contain 7,312,498 orders, represented by 10,631,250 product-line records, from 779,744 consumers at 260 stores in a major city. For each order, we observe the consumer, purchase date, store, and products

purchased. Because consumers can purchase from any store in the chain, our behavioral outcomes aggregate purchases chain-wide rather than only at the focal store.

Temporary store closures during COVID-19-related lockdowns provide the access shocks¹. We identify 18 closures lasting 10 to 29 days after applying the screens described in Appendix A.1. Our results are robust to a more relaxed selection rule leading to 22 eligible closures (Appendix Table A1). Panel A of Figure 1 illustrates the event-time construction. For each closure event e , we construct four pre-closure windows, indexed by -4 through -1 , and four post-reopening windows, indexed by 1 through 4 . Each window has the same length L_e as the closure. Period 0 corresponds to the interval during which the focal store is closed. Our main analysis compares behavior before closure with behavior after reopening and excludes period 0. This construction allows outcomes to be measured over comparable windows within each event.

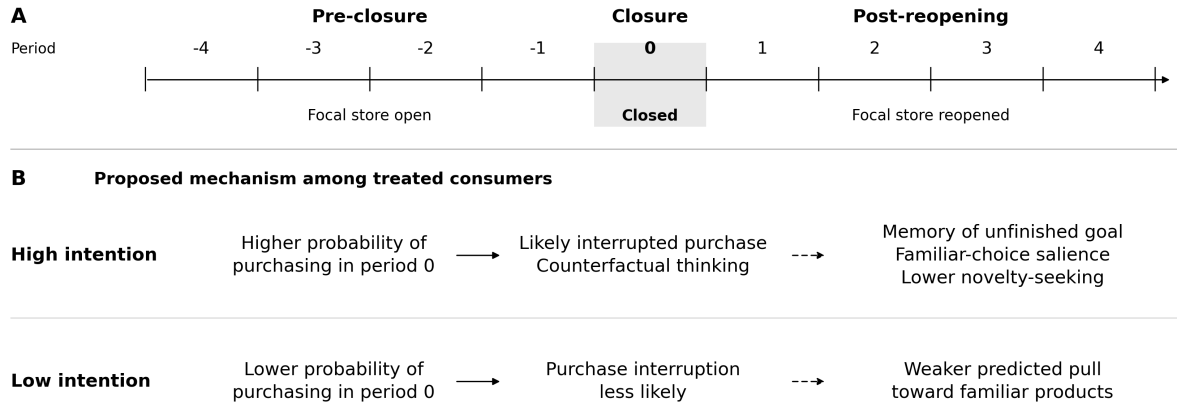


Figure 1: Event timing and the proposed mechanism

We assign treatment exposure through consumers’ pre-closure store relationships. A consumer can purchase at multiple stores, so a store closure is most likely to represent a meaningful access shock when the closing store was central to the consumer’s prior purchases. We therefore restrict attention to consumers with at least five pre-closure purchase days and define a consumer’s focal store as the store accounting for at least 80% of her pre-closure purchases. Treated consumers have a focal store that closes; control consumers have a focal store among five matched stores that remain open. For each closing store, we select without replacement the five eligible control stores with the closest setup dates. Because the firm

¹Contemporaneous reports illustrate that local containment measures in major Chinese cities included the closure of neighborhood stores and shopping malls; see, for example, https://www.xinhuanet.com/2021-01/19/c_1127001704.htm and <https://m.bjnews.com.cn/detail/162795097414850.html>.

opened stores in rollout cohorts, setup timing proxies for stores’ operational status and strategic importance. We exclude university and college stores because campus lockdowns and summer departures sent many students outside the focal city, where their purchases are not observed; zero local transactions could therefore reflect population departure rather than a supply interruption. Treatment therefore captures exposure to a localized access restriction, while subsequent outcomes continue to include purchases at any store in the observed chain. The resulting sample contains 40,148 consumers, including 7,390 treated and 32,758 control consumers.

3.1 Identifying Likely Purchase Interruption

Panel B of Figure 1 illustrates why the same closure can have different consequences for consumers. A closure is more likely to interrupt an intended purchase for a consumer who would otherwise have purchased during period 0. Under the proposed mechanism, an interrupted purchase associated with an established routine can make the familiar purchase path salient through counterfactual thinking and the continued accessibility of the unfinished goal. This representation can carry into product choices after reopening. Consumers who were less likely to purchase during period 0 are correspondingly less likely to experience this interruption. The high- and low-intention labels in the figure refer to these differences in predicted purchase propensity, which we construct using only pre-closure information.

We infer this latent purchase propensity using an XGBoost classifier trained exclusively on pre-closure information. The prediction target is whether a consumer makes at least one purchase at any store in the chain during a window of the same length as the focal closure. Thus, treatment is assigned through the preferred pre-closure store, whereas the classifier target and behavioral outcomes are measured chain-wide. Because the probability of at least one purchase mechanically rises with window length, we estimate a separate model for each observed closure duration. Training uses relative pre-periods -4 through -2 , with predictors constructed from information available before each prediction window; relative period -1 is held out for validation. We also conduct an additional period-0 audit among control consumers, whose matched focal stores remain open and whose realized purchase behavior is therefore observable during the focal closure window. For treated consumers, the corresponding period-0 purchase outcome absent the closure is counterfactual, which is precisely why the predicted propensity is needed.

The predictors summarize the consumer’s purchase history before the prediction window: recent purchase frequency, recency since the last purchase, total purchase history, focal-store loyalty, basket com-

position, discount and coupon use, day-of-week purchasing patterns, demographics, and push-notification history. The classifier produces a predicted probability between zero and one. We define $\text{High} = 1$ when this predicted probability is at least 0.5 and $\text{High} = 0$ otherwise. The indicator is constructed ex ante for both treated and control consumers; a likely interrupted purchase occurs when a consumer is both treated and high predicted intention.

Table 1: Purchase-intention classifier performance on the final cohort

Evaluation sample	N	Prevalence	ROC-AUC	PR-AUC	Brier	Calibration slope
Control, period 0	32,758	.263	.781	.583	.172	.471
Control, pre-period -1	32,758	.308	.814	.677	.164	.536
Treated, pre-period -1	7,390	.314	.818	.697	.162	.537

Notes: All rows use the exact final 18-event cohort. Prevalence is the observed share purchasing at least once in the evaluation window. ROC-AUC measures ranking performance; PR-AUC should be compared with prevalence, the performance of an uninformative ranking. The Brier score is the mean squared probability error, for which lower values are better. The corresponding constant-prevalence Brier scores are .194, .213 and .216. Perfect calibration has intercept zero and slope one; the estimated intercepts are $-.599$, $-.309$ and $-.276$, respectively. Period-0 outcomes are not observable for treated consumers absent the closure, so that validation is conducted only among controls.

Table 1 shows that the model meaningfully ranks likely purchasers before the focal post-reopening outcomes are observed. The period-0 control audit is especially informative because it evaluates the prediction in the same relative window in which treated consumers face the closure. Its ROC-AUC is 0.781 and its PR-AUC is 0.583, compared with an observed purchase prevalence of 0.263. Its Brier score is 0.172, below the 0.194 score obtained by assigning every consumer the sample purchase rate. The purpose of the classifier in the design is therefore not to recover a perfectly observed psychological intention; it is to separate consumers with materially different ex-ante propensities to purchase during the window in which the access shock occurs. Appendix A.4 reports confusion matrices, calibration, duration-specific performance, cutoff tradeoffs, and score overlap.

3.2 Analysis Sample and Raw Closure Shock

For the descriptive access-shock evidence in this section, we measure purchase frequency as the number of days with at least one purchase divided by the number of calendar days L_e in the event window. This normalization makes purchase activity comparable across closure events of different durations.

The event construction yields 321,184 consumer-period observations for purchase activity. Novelty-seeking is defined only in windows containing at least one purchase. There are 105,854 such purchasing

windows; after the consumer fixed-effects estimator removes consumers observed in only one product-choice period, the main novelty-seeking regressions use 99,644 observations. The classifier places 2,041 treated consumers, or 27.6%, in the high-predicted-intention group.

Table 2: Four-cell descriptive statistics before closure

Variable	Low predicted intention		High predicted intention	
	Control	Treated	Control	Treated
Consumers	23,934	5,349	8,824	2,041
Predicted purchase-intention probability	.135 (.141)	.132 (.139)	.773 (.157)	.785 (.158)
Pre-period purchase frequency	.022 (.035)	.021 (.033)	.127 (.116)	.114 (.112)
Pre-period novelty-seeking	.604 (.392)	.602 (.392)	.450 (.309)	.498 (.305)
Consumers with at least one pre-period purchase	12,317	2,745	8,798	2,034
Share of pre-period windows with a purchase	.214	.211	.751	.753
Closure duration in days	16.2 (5.6)	16.9 (6.5)	18.8 (6.4)	21.7 (7.2)

Notes: The unit is a consumer. Standard deviations are in parentheses. Pre-period purchase frequency is first averaged over relative periods -4 through -1 within a consumer. Pre-period novelty-seeking is averaged over purchasing windows and is reported for consumers with at least one pre-period purchase. The purchase share is the fraction of the four pre-period windows in which the consumer purchases at least one product. High and low groups differ mechanically in purchase propensity; the relevant descriptive comparison is treated versus control within each predicted-intention group. Closure duration is repeated at the consumer level and is shown descriptively, without consumer-level balance inference.

Table 2 displays the four cells that form the triple-differences comparison. The classifier creates economically distinct intention groups: high-intention consumers make at least one purchase in about three quarters of their pre-period windows, compared with about one fifth for low-intention consumers. At the same time, treated and control consumers have very similar predicted purchase propensities within each intention stratum: 0.132 versus 0.135 in the low group and 0.785 versus 0.773 in the high group. The table also makes clear why the design is based on changes in treated-control gaps rather than equality of pre-period levels. Consumer fixed effects absorb time-invariant level differences, while the pre-period event-study tests below evaluate whether the treated-control gap was already evolving differently between the high- and low-intention groups before the closure.

The closures generate a sharp first-stage change in access. Figure 2 plots purchase activity around the closure window. Daily purchase frequency among treated consumers falls from 0.039 in period -1 to 0.007 during the closure, an 82% decline, while the control mean is 0.036 during the same relative period. In the first post-reopening window, treated and control means are both 0.039. The raw paths therefore show that

the focal closures sharply restrict treated consumers' purchasing opportunities and that access is restored after reopening. Because purchase frequency is measured across the entire observed chain, the drop also shows that substitution to other chain stores does not undo the access shock during the closure window.

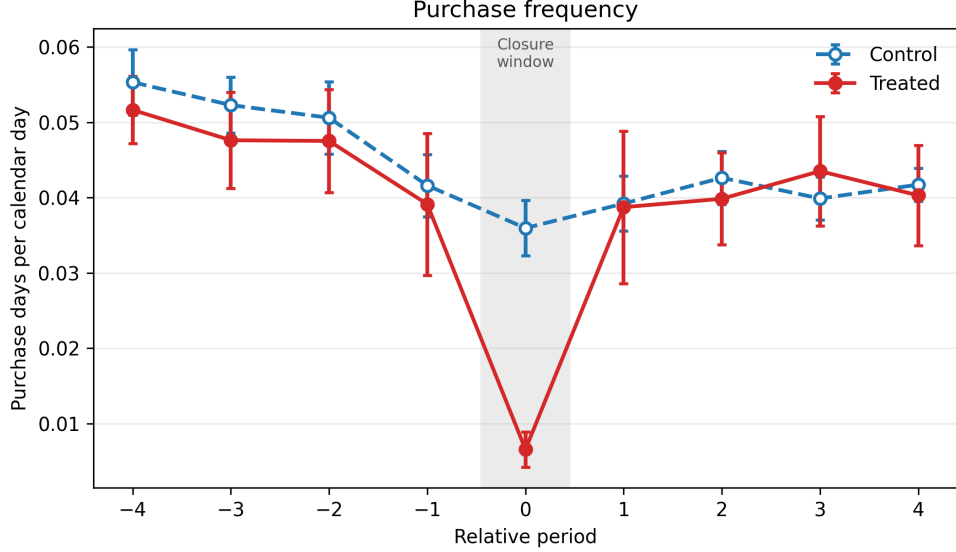


Figure 2: Store closures sharply reduce treated purchase frequency during the closure window

Notes: The figure plots consumer-weighted pooled means for the 18-closure analysis sample; each relative period contains 32,758 control and 7,390 treated consumers. Bars are 95% confidence intervals. Relative period 0 is the closure window. Purchase frequency is the number of days divided by the number of calendar days in the event-length window.

4 Empirical Strategy and Main Results

4.1 Measuring Novelty-Seeking

Let i index consumers, e closure events, t relative periods, and let L_e denote the duration in days of closure event e . For each event, relative periods $t = -4, \dots, 4$ have length L_e , with period 0 corresponding to the closure window. The estimation sample omits period 0 and compares four pre-closure windows with four post-reopening windows.

Our primary outcome measures exploration relative to the consumer's own purchase history. Let $\mathcal{J}_{iet}$ denote the set of distinct products that consumer i purchases during event window t , and let $\mathcal{H}_{iet}^-$ denote the set of products the consumer purchased before that window. We define novelty-seeking as

$$Y_{iet}^N = \frac{|\mathcal{J}_{iet} \setminus \mathcal{H}_{iet}^-|}{|\mathcal{J}_{iet}|}. \quad (6)$$

The set difference $\mathcal{J}_{iet} \setminus \mathcal{H}_{iet}^-$ contains the products purchased in the current window that do not appear in the consumer’s prior purchase history, and $|\cdot|$ denotes the number of distinct products in a set. The outcome therefore ranges from zero to one. For example, if a consumer buys three distinct products in a window and one has never been purchased before, novelty-seeking equals $1/3$. A higher value means that a larger share of the consumer’s current product set is unfamiliar based on her own observed history.

Novelty-seeking is defined only when the consumer purchases at least one product in the window, because otherwise the denominator in equation (6) is zero. The estimand concerns the *composition of product choice conditional on purchasing*, rather than the probability of making a purchase. Accordingly, our estimates capture how purchase interruption changes the composition of realized product choices, and not its unconditional effect on the probability of exploring a novel product. Section 6.2 complements this consumer-specific measure with new-product-seeking, which instead classifies novelty according to whether a product has recently appeared in the chain.

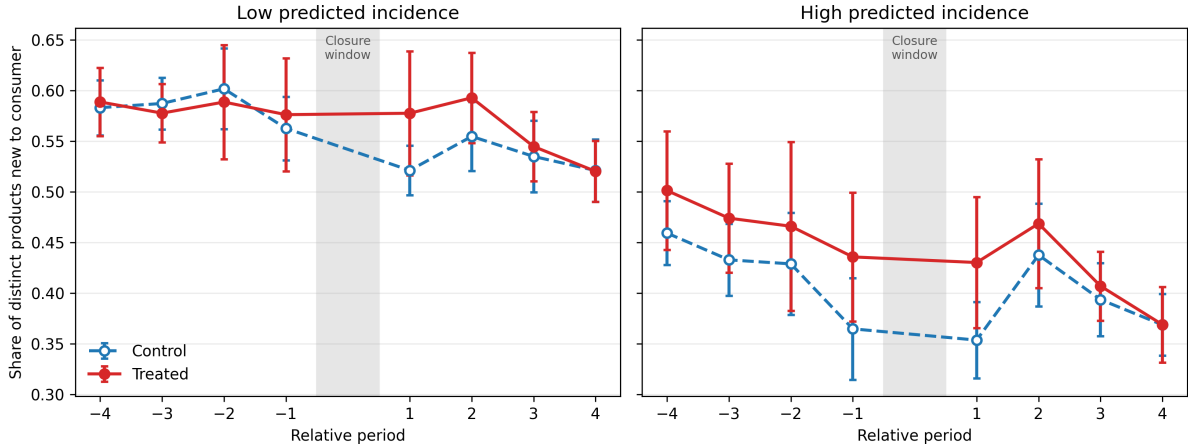


Figure 3: Raw novelty-seeking paths motivate the high-minus-low comparison

Notes: The figure plots consumer-weighted pooled means for the 18-closure analysis sample. Bars are 95% confidence intervals. Across the eight plotted periods, observed-window counts range from 4,227 to 6,354 for control-low, 919 to 1,397 for treated-low, 4,918 to 7,469 for control-high and 1,171 to 1,714 for treated-high consumers. Relative period 0, the closure window, is omitted.

Figure 3 plots the raw novelty-seeking paths for the four cells underlying the triple-differences design. In the low-predicted-intention group, treated and control means are close before the closure and the treated-control gap becomes more positive after reopening. In the high-predicted-intention group, the post-reopening change in the treated-control gap is much smaller. The empirical object of interest is precisely this difference between the two treated-control responses, which motivates the high-minus-low comparison estimated below.

Because novelty-seeking is observed only in purchasing windows, Figure 4 plots the probability that a consumer makes at least one purchase in each relative period. This probability is also the probability that the consumer-period enters the novelty-seeking sample. Within each predicted-intention stratum, treated and control purchase-probability paths track closely around reopening. Thus, the product-choice result is not accompanied by a comparably large treated-control divergence in the extensive margin within either intention group. The absence of a large treated-control divergence in purchase probability reduces concerns that the novelty result is mechanically driven by differential sample entry, although it does not eliminate selection among consumers who purchase. We therefore retain a conditional-on-purchase interpretation throughout.

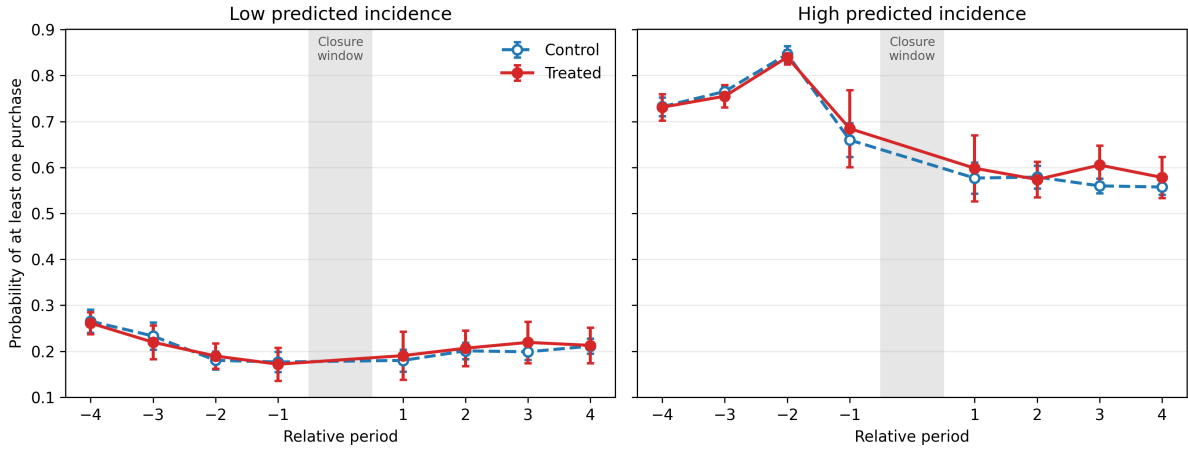


Figure 4: Purchase probability evolves similarly for treated and control consumers within predicted-intention groups

Notes: The figure plots the share of consumer-period observations containing at least one purchase. This is also the share of windows in which novelty-seeking is defined. Each relative period contains 23,934 control-low, 5,349 treated-low, 8,824 control-high and 2,041 treated-high consumers. Bars are 95% confidence intervals. Means are consumer weighted. Relative period 0 is omitted to match the main regressions.

4.2 Triple Difference-in-Differences

Figure 1 illustrates two dimensions of our empirical comparison: behavior before closure versus after reopening in Panel A, and high versus low predicted purchase intention in Panel B. We construct both intention groups among treated consumers and consumers associated with matched stores that remain open. Comparing treated and control consumers provides the third dimension. The resulting triple difference asks whether the treated-control change in novelty-seeking is more negative among consumers whose purchases were more likely to have been interrupted. For intention group $g \in \{L, H\}$, let Δ_g denote the treated-control

difference-in-differences:

$$\Delta_g = \left(\bar{Y}_{T,g}^{Post} - \bar{Y}_{T,g}^{Pre} \right) - \left(\bar{Y}_{C,g}^{Post} - \bar{Y}_{C,g}^{Pre} \right). \quad (7)$$

Δ_L measures the treated-control change in novelty-seeking among consumers with low predicted purchase intention, and Δ_H measures the corresponding change among consumers with high predicted purchase intention. Both groups experience closure exposure, while the likelihood that an intended purchase is interrupted is higher in the high-intention group. The triple difference compares these two responses:

$$DDD = \Delta_H - \Delta_L. \quad (8)$$

Under the identifying conditions described below, the third difference removes the component of the closure response shared across intention groups and identifies the incremental response associated with a higher likelihood of purchase interruption. This logic follows the canonical triple-differences design (Gruber, 1994) and is closely related to recent work using temporary disruptions to distinguish exposed consumers by the likelihood that a choice opportunity was displaced (Levine and Hristakeva, 2026).

We implement this comparison in the pooled regression

$$Y_{iet}^N = \delta^B \cdot \text{Post}_{et} \times \text{Treated}_{ie} + \beta \cdot \text{Post}_{et} \times \text{High}_{ie} + \delta^D \cdot \text{Post}_{et} \times \text{Treated}_{ie} \times \text{High}_{ie} + \phi_{ie} + \omega_t + \gamma_m + \epsilon_{iet}, \quad (9)$$

where Y_{iet}^N is novelty-seeking from equation (6). Post_{et} equals one in relative periods 1 through 4 and zero in relative periods -4 through -1 . Treated_{ie} equals one when consumer i is assigned to the store that closes in event e , and High_{ie} equals one when the ex-ante predicted purchase probability during the closure window is at least 0.5. Only the treated-high cell combines closure exposure with a high likelihood that an intended purchase was blocked.

The fixed effects serve distinct roles. Consumer fixed effects ϕ_{ie} absorb all characteristics that are constant within a consumer, including the levels of Treated, High, and their interaction. Relative-period fixed effects ω_t absorb common movement by event time, while calendar-month fixed effects γ_m absorb calendar shocks shared by consumers observed in the same month, such as chain-wide changes in demand or product introductions. The remaining error is ϵ_{iet} . Because the access shock is shared by consumers within a closure event, standard errors are clustered by closure event.

The coefficients map directly to equations (7)–(8). For low-intention consumers, $\text{High} = 0$, so the

treated-control post response is $\delta^B = \Delta_L$. For high-intention consumers, the treated-control response is $\delta^B + \delta^D = \Delta_H$. Hence $\delta^D = \Delta_H - \Delta_L$ is the focal DDD. The coefficient β captures a high-minus-low post shift that is common to treated and control consumers; it is not a closure effect. This decomposition is important for interpreting the estimates below: the causal object of interest is not the high-intention coefficient in isolation, but how the high-intention closure response differs from the low-intention closure response.

4.3 Identification

The interpretation of δ^D as the incremental response associated with likely purchase interruption rests on two conditions. First, other consequences of closure exposure should generate similar treated-control changes among high- and low-intention consumers. These consequences may include changes in product availability, reopening promotions, shopping convenience, and consumption occasions. The third difference removes these components when their effects are shared across intention groups. The relevant concern is therefore a closure-related change that affects the two intention groups differently. For example, relatively greater promotion of familiar products to high-intention consumers, or lower exposure to novel products when they return, could also generate a negative novelty-seeking DDD.

Second, absent the closure, the treated-control gap for high-intention consumers should have evolved like the treated-control gap for low-intention consumers. This is the parallel-triple-trends condition. Importantly, it concerns changes in the treated-control gaps rather than equality of pre-period levels. Consumer fixed effects absorb persistent level differences; identification requires that the *difference between the two treated-control gaps* was not already trending before the closure.

We evaluate this condition using an event-study version of equation (9). Relative period -1 , the last pre-closure window, is the omitted reference. For each earlier pre-period $t \in \{-4, -3, -2\}$, we estimate the high-minus-low change in the treated-control gap relative to period -1 , and then test whether the three pre-period DDD coefficients are jointly zero. For novelty-seeking, the three lead estimates range from 1.60 to 2.64 percentage points, every 95% confidence interval includes zero, and the joint test yields $p = 0.841$. The pre-period dynamics therefore provide no evidence that the high-minus-low treated-control gap was already moving before the focal closure. Appendix B reports the full event-study coefficients and additional diagnostics.

4.4 Main Results

Table 3 reports the group-specific closure responses and their difference. Reading the table from top to bottom is useful because the DDD is constructed from the first two rows. Among low-predicted-intention consumers, the treated-control post-reopening response is a 3.13-percentage-point increase in novelty-seeking ($SE = 1.78$; $p = 0.098$). Among high-predicted-intention consumers, the corresponding response is a 1.02-percentage-point decline ($SE = 0.72$; $p = 0.175$). The difference between these responses is -4.15 percentage points ($SE = 1.84$; $p = 0.038$), which is the focal estimate δ^D .

Table 3: Purchase interruption narrows subsequent novelty-seeking

Effect	Novelty-seeking
Low predicted purchase intention (δ^B)	.0313* (.0178) [.0976]
High predicted purchase intention ($\delta^B + \delta^D$)	-.0102 (.0072) [.1745]
Incremental interruption effect: High minus low (δ^D)	-.0415** (.0184) [.0380]
Pre-period outcome mean	.4978
Observations	99,644
R^2	.416
Within R^2	.001

Notes: Novelty-seeking is the share of distinct products purchased in a window that the consumer had not purchased before that window. The low- and high-intention rows report treated-control post contrasts within each predicted-intention group. The bold row is their difference, the focal triple-differences estimand. Standard errors, clustered by closure event (18 clusters), are in parentheses; p-values are in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. All regressions include consumer, relative-period, and calendar-month fixed effects and omit the closure window.

The positive 3.13-point response among low-intention consumers is informative about the general effect of closure exposure and about why the third difference is needed. Novelty is defined relative to each consumer's accumulated purchase history. During the closure, control consumers continue to purchase and can move products from the "untried" set into their familiar-product histories, while treated consumers have fewer opportunities to do so. Products introduced during the closure may likewise be tried by controls before treated consumers regain access. After reopening, a product can therefore remain classified as novel for a treated consumer even when it is already familiar to a comparable control consumer. These forces naturally push the low-intention treated-control contrast upward and represent consequences of the closure

that do not require an intended purchase to have been interrupted. Importantly, Section 6.2 shows that the high-minus-low differential remains nearly identical when exploration is defined using recently introduced products rather than the consumer’s accumulated purchase history.

The high-intention response is markedly different. For consumers whose purchase propensity during the closure was high, the treated-control contrast is 4.15 percentage points lower than the low-intention contrast. In other words, making the closure much more likely to bind against an intended purchase more than offsets the positive exploration response associated with closure exposure alone. The focal result is this *difference between closure responses*, not the -1.02 -point high-intention coefficient by itself. The DDD asks whether the high-intention response is different from the counterfactual closure response represented by the low-intention group, and the answer is a statistically significant -4.15 -point shift.

The magnitude is economically meaningful. Relative to the pre-period mean novelty-seeking of 49.78%, the 4.15-point DDD represents an 8.3% reduction. Put differently, among purchasing windows, likely purchase interruption shifts the composition of subsequent purchases toward products already present in the consumer’s history. Under the identifying conditions in Section 4.3, this estimate supports Prediction 1: an intended but unrealized purchase narrows subsequent product exploration.

This interpretation also clarifies what the DDD contributes beyond a standard closure comparison. An average treated-control estimate would combine consumers for whom the closure merely changed access with consumers for whom it likely blocked an active purchase. The high-minus-low comparison separates these two components and reveals that the behavioral response associated with likely interruption runs in the opposite direction from the general closure-exposure response. Section 5 next tests whether the cross-consumer pattern follows the heterogeneity predicted by familiar-choice salience and examines two concrete alternative explanations. Section 6 asks whether the same result appears when exploration is defined by adoption of products newly introduced by the firm.

5 Evidence on Behavior Mechanism and Alternative Explanations

Prediction 2 provides a mechanism-linked test of familiar-choice salience. If an interrupted purchase reactivates an established consumption path, the reduction in exploration should be strongest among consumers whose prior choices are already more concentrated on familiar products. We test this implication using baseline novelty-seeking measured before the focal closure. The analysis asks whether the high-minus-

low novelty-seeking DDD becomes more negative as consumers' prior choices become more concentrated on familiar products.

5.1 Heterogeneity by Baseline Novelty-Seeking

For each consumer, we calculate novelty-seeking separately in relative periods -8 through -1 and then average across periods in which the consumer purchases. Each of these periods has the same length as the associated closure event. Let O_{ie} denote the number of distinct orders consumer i places over these eight baseline windows. We require $O_{ie} \geq 5$ so that the baseline measure is based on an established purchase history rather than one or two isolated transactions. The raw baseline novelty measure ranges from zero, when all purchased products are familiar, to one, when all distinct products purchased in each purchasing window are new relative to the consumer's earlier history.

The resulting sample contains 18,525 eligible consumers. Baseline novelty varies broadly: the mean is 0.508, the standard deviation is 0.269, the median is 0.500, and the 10th and 90th percentiles are 0.167 and 0.889. Figure 5 shows that the heterogeneity test therefore uses variation across a wide range of exploratory behavior rather than a narrow split around a single cutoff.

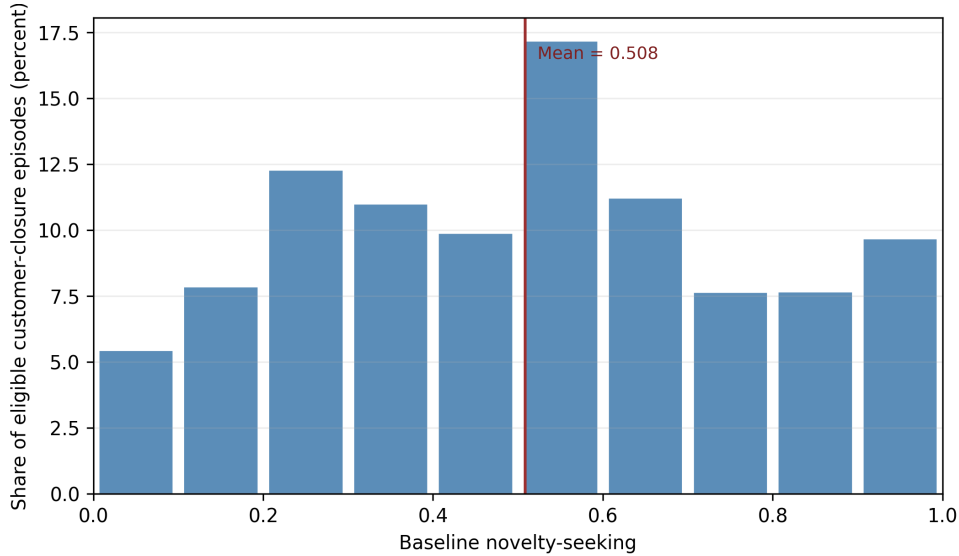


Figure 5: Distribution of baseline novelty-seeking

Notes: The figure shows ten equal-width bins of the raw consumer-level baseline novelty measure. The measure is the mean of within-period novelty-seeking over purchasing periods from relative period -8 through -1 . Eligible consumers have at least five orders across these eight closure-length periods. The last bin includes one. The vertical line marks the sample mean. The sample contains 18,525 eligible consumers.

Let N_{ie} denote standardized baseline novelty-seeking. We first estimate a direct baseline-novelty specification that augments the main DDD with a complete Post-by-baseline-novelty interaction hierarchy:

$$\begin{aligned}
Y_{iet}^N = & \delta^B(\text{Post}_{et} \times \text{Treated}_{ie}) + \beta(\text{Post}_{et} \times \text{High}_{ie}) + \delta^D(\text{Post}_{et} \times \text{Treated}_{ie} \times \text{High}_{ie}) \\
& + \text{Post}_{et} \times N_{ie} (\eta_0 + \eta_T \text{Treated}_{ie} + \eta_H \text{High}_{ie} + \theta^D \text{Treated}_{ie} \times \text{High}_{ie}) \\
& + \phi_{ie} + \omega_t + \gamma_m + \varepsilon_{iet}.
\end{aligned} \tag{10}$$

The coefficient θ^D measures how the high-minus-low novelty-seeking DDD changes with a one-standard-deviation increase in baseline novelty. Prediction 2 implies $\theta^D > 0$: the negative interruption response should attenuate as consumers become more exploratory.

Because novelty is defined relative to a consumer's accumulated purchase history, the measured baseline novelty share can also depend on the amount of history available. As consumers accumulate orders, more products enter their familiar set, so the same underlying willingness to explore can generate a lower measured novelty share when the observed history is longer. In the eligible sample, baseline novelty is correlated at -0.329 with $\log(1 + O_{ie})$: mean baseline novelty is 0.620 among consumers with exactly five baseline orders and 0.365 among consumers with at least 20. We therefore report a second specification that adds an analogous complete Post-interaction hierarchy for

$$Q_{ie} = \frac{\log(1 + O_{ie}) - \overline{\log(1 + O)}}{\text{SD}[\log(1 + O)]},$$

the standardized log number of baseline orders. This order-history-adjusted specification allows the DDD to vary separately with baseline novelty and with the amount of purchase history used to construct it. Because N_{ie} , Q_{ie} , treatment status, and predicted-intention status are time invariant within a consumer, their levels are absorbed by ϕ_{ie} ; only their interactions with Post are estimable.

Table 4 reports both specifications. Column (1) directly interacts the DDD with baseline novelty-seeking. The estimated gradient is positive at 2.29 percentage points per standard deviation ($\text{SE} = 1.43$; one-sided $p = .064$). Column (2) additionally allows the post-period response in every DDD cell to vary with baseline order history. The estimated baseline-novelty gradient increases modestly to 2.94 percentage points ($\text{SE} = 1.47$; one-sided $p = .031$). The two estimates have the same predicted sign and similar magnitudes, indicating that the heterogeneity pattern is present both before and after separating baseline novelty

Table 4: Baseline novelty-seeking attenuates the interruption effect

	(1) Baseline-novelty interaction	(2) Order-history-adjusted interaction
DDD at mean covariate values (δ^D)	-3.05 (2.01)	-1.30 (2.00)
Change in DDD per SD of baseline novelty (θ^D)	2.29* (1.43)	2.94** (1.47)
One-sided p -value for $\theta^D > 0$	.0636	.0312
Joint pretrend p -value for θ^D	.4094	.5611
Observations	79,578	79,578
Closure-event clusters	18	18
Consumer, relative-period, and calendar-month fixed effects	Yes	Yes
Post-by-baseline-order interaction hierarchy	No	Yes

Notes: Estimates and standard errors are in percentage points. Baseline novelty-seeking is the consumer-level mean novelty over relative periods -8 through -1 and is standardized over eligible consumers. Column (1) estimates equation (10). Column (2) additionally includes a complete Post-by-baseline-order interaction hierarchy using standardized $\log(1 + \text{baseline orders})$. The DDD is evaluated at mean baseline novelty in column (1) and at mean baseline novelty and mean baseline order history in column (2). Both specifications use the restricted heterogeneity sample. Standard errors use CRV1 clustering by the 18 closure events. * $p < 0.10$, ** $p < 0.05$, based on the one-sided directional test of $\theta^D > 0$.

from the amount of purchase history used to construct it.

Across the two specifications, a one-standard-deviation increase in baseline novelty makes the interruption-related DDD 2.29 to 2.94 percentage points less negative. The decline in exploration is therefore concentrated among consumers whose pre-closure choices were more heavily oriented toward familiar products, as predicted by familiar-choice salience. The corresponding event-study tests do not reject a differential pretrend in either specification ($p = .409$ and $p = .561$). Appendix B.2 reports the dynamic estimates for the order-history-adjusted specification.

5.2 Alternative Explanations

We next examine two exposure-based explanations that could generate lower post-reopening exploration without the familiar-choice channel: differential exposure to new-product marketing and differential access to novel products after reopening. Each explanation requires a specific observable pattern in the data, allowing us to evaluate whether it can account for the sign and magnitude of the novelty DDD.

5.2.1 New-Product Notifications

The digital app sends push notifications to consumers based on their purchase histories, including campaigns that specifically promote newly introduced products. Consumers typically enter a “dormant” stage after 30 days without a purchase, and the firm’s history-based notification algorithm determines subsequent messaging. This creates a plausible alternative explanation because the same purchase histories that predict intention can also affect notification exposure. If treated high-intention consumers systematically received fewer new-product messages after the closure, their lower exploration could reflect lower marketing exposure rather than a shift toward familiar choices.

To evaluate this channel, we count unique new-product campaigns received per consumer-day and estimate the same treated-control, high-minus-low comparison used for the main outcome. The sign required by the alternative explanation is negative: relative to the corresponding controls, treated high-intention consumers would need to receive fewer new-product campaigns than treated low-intention consumers. Instead, the estimated contrast is positive. During the closure, the relative high-minus-low exposure change is +0.0041 campaigns per consumer-day (95% CI: 0.0019 to 0.0064; restricted wild-cluster $p = 0.003$). After reopening, it remains positive at +0.0017 campaigns per day (95% CI: 0.0001 to 0.0032; $p = 0.047$), and the event-study leads do not reject a common pretrend ($p = 0.500$).

The direction is informative. Treated high-intention consumers are targeted with *more*, not fewer, new-product campaigns relative to the DDD comparison group. Although the records do not reveal whether consumers viewed these messages, the observed targeting differential runs in the opposite direction from that required by the notification-based explanation. If new-product notifications increase awareness or trial, this differential exposure would tend to raise their exploration and therefore work against the negative novelty DDD. The targeting records therefore do not support a relative reduction in new-product campaign exposure as an explanation for the negative novelty-seeking DDD. Appendix C.1 reports the campaign construction, alternative exposure measures, and complete estimates.

5.2.2 Reopening Assortments

A second explanation is that high- and low-intention consumers encounter different opportunities to choose novel products after reopening. This can occur if the store assortment changes over time and the two groups return at different times. We therefore separate the argument into two steps: whether realized

assortments evolve after reopening, and whether the groups' actual return timing translates that evolution into meaningfully different exposure to products that are novel for each consumer.

We first characterize realized assortment at treated stores over four fixed seven-day periods after reopening. We use three complementary measures. *Core-product coverage* is the share of regularly sold pre-closure products that continue to appear in a given week. *Products per 50 baskets* measures the number of distinct products sold in standardized samples of 50 consumer-day baskets, which removes the mechanical effect of different transaction volumes. *Pre-menu overlap* measures the similarity between products sold in a post-reopening week and the store's pre-closure product set. Core-product coverage does not change significantly across the four weeks ($p = 0.382$ jointly), but standardized product breadth and pre-menu overlap do evolve ($p = 0.011$ and $p = 0.018$ jointly). Thus, assortment evolution is meaningful enough that return timing is worth evaluating directly rather than assuming that all returners face the same menu environment.

High-intention consumers do return earlier. Among treated consumers who return to the focal store within 28 days, high-intention consumers return 2.87 days earlier on average ($SE = 0.29$); 53.6% return in week 1 compared with 36.9% of low-intention returners. To translate this timing difference into consumer-specific opportunity, we construct, for every returner and each possible return week, a leave-one-consumer-out realized product set. Excluding the focal consumer prevents her own purchase from mechanically affecting the measured assortment. We then calculate the share and count of products in that set that the consumer had never purchased by reopening. Finally, we subtract each consumer's own four-week average opportunity from the opportunity in the week in which she actually returns. This isolates the component of novel-product exposure generated by return timing rather than stable differences across consumers.

The high-minus-low timing-induced difference in the novel-product share is -0.10 percentage points ($SE = 0.06$; 95% CI: -0.22 to $+0.02$). The corresponding difference in the number of novel products is -0.28 products ($SE = 0.38$; 95% CI: -1.09 to $+0.54$). Thus, although realized assortments evolve and high-intention consumers return earlier, their return timing generates only a small difference in measured novel-product opportunity. These results provide little support for a substantial return-timing-driven gap in realized assortment exposure among consumers returning to the focal store within 28 days. [Appendix C.2](#) reports the weekly assortment paths, construction details, and matched-store comparisons.

6 Robustness Checks

We conduct four sets of robustness analyses around the main novelty-seeking result. First, the main specification classifies consumers into high- and low-predicted-intention groups using a cutoff of 0.5. We replace this binary classification with the continuous predicted purchase probability to examine whether the result depends on the particular threshold. Second, our main novelty measure defines a product as novel when the consumer has not purchased it before. We use an alternative measure based on products that have recently appeared in the chain to examine whether purchase interruption also affects new-product adoption. Third, because a substantial share of purchases in our setting involves coffee, we re-estimate the novelty-seeking specification using only non-coffee products. Finally, we account for differences in the local competitive environment surrounding consumers' preferred stores and for broader local post-reopening shocks.

6.1 Continuous Predicted Purchase Probability

Our main specification converts the predicted purchase probability into a binary indicator, High_{ie} , using a cutoff of 0.5. Although this classification gives the DDD a transparent interpretation, consumers whose predicted probabilities are 0.49 and 0.51 are assigned to different groups despite having similar predicted purchase intentions. We therefore examine whether the main result also appears when we use the full variation in predicted purchase probability.

Let $s_{ie} \in [0, 1]$ denote the predicted probability that consumer i would make at least one purchase during the closure window associated with event e , and let

$$\tilde{s}_{ie} = s_{ie} - \bar{s}$$

denote the score centered at its estimation-sample mean. Replacing High_{ie} in equation (9) with $\tilde{s}_{ie}$ gives

$$Y_{iet}^N = \delta_0^B (\text{Post}_{et} \times \text{Treated}_{ie}) + \beta^S (\text{Post}_{et} \times \tilde{s}_{ie}) + \delta^S (\text{Post}_{et} \times \text{Treated}_{ie} \times \tilde{s}_{ie}) + \phi_{ie} + \omega_t + \gamma_m + \epsilon_{iet}. \quad (11)$$

Because the score is centered, δ_0^B measures the treated-control post-reopening response for a consumer with the average predicted purchase probability. The coefficient β^S captures how the post-period outcome varies with predicted purchase intention in a way common to treated and control consumers. The coefficient

of interest is δ^S : it measures how the treated-control post-reopening response changes as the predicted probability that the closure interrupted a purchase increases. A negative δ^S therefore corresponds directly to the prediction that consumers whose purchases were more likely to have been interrupted subsequently exhibit a larger reduction in novelty-seeking.

Using novelty-seeking as the dependent variable, we estimate $\delta^S = -0.0452$ (SE = 0.0276; $p = 0.120$). Since the predicted-intention score ranges from zero to one, this coefficient implies that a 10-percentage-point increase in predicted purchase intention is associated with a 0.45-percentage-point more negative treated-control post-reopening response in novelty-seeking. The continuous specification therefore preserves the negative relationship between predicted purchase intention and the treated-control post-reopening response found in the binary specification, although the interaction is not statistically distinguishable from zero. Appendix Table A14 reports the complete specification and corresponding diagnostics.

6.2 Exploration of Recently Introduced Products

Our main novelty-seeking measure focuses on whether consumers depart from their own familiar purchase histories. An alternative perspective, especially relevant for firms introducing new products, is whether consumers adopt products that have recently appeared in the marketplace. The two measures capture related but distinct forms of exploration. A product may be new to a particular consumer even if it has been sold by the firm for a long time, whereas a recently introduced product represents an innovation from the firm's perspective.

Let F_j denote the first date on which product j appears in the chain's transaction records, and let $\mathcal{W}_{et}$ and $\mathcal{W}_{e,t-1}$ denote the current and immediately preceding event windows. We define new-product-seeking as

$$Y_{iet}^{NP} = \frac{|\{j \in \mathcal{J}_{iet} : F_j \in \mathcal{W}_{e,t-1} \cup \mathcal{W}_{et}\}|}{|\mathcal{J}_{iet}|}. \quad (12)$$

Thus, Y_{iet}^{NP} is the share of distinct products purchased in the current window whose first observed chain sale occurs in either the current or immediately preceding event window. Like the main novelty measure, it is defined conditional on the consumer making a purchase. The first observed sale serves as our empirical measure of when a product newly appears in the chain.

We first apply the binary DDD specification in equation (9). Among low-predicted-intention consumers, the treated-control post-reopening response in new-product-seeking is +0.80 percentage points (SE = 1.13;

$p = 0.487$). Among high-predicted-intention consumers, the corresponding response is -3.37 percentage points ($SE = 1.45$; $p = 0.033$). Their difference is -4.17 percentage points ($SE = 0.89$; $p < 0.001$), equivalent to 28.6% of the 14.56% pre-period mean. The high-minus-low pretrend test does not reject parallel trends ($p = 0.114$).

This result strengthens the interpretation of the main finding in two ways. First, the -4.17 -point DDD is almost identical in absolute magnitude to the -4.15 -point estimate obtained when novelty is defined relative to the consumer’s own history. Second, the treated-control post-reopening change among high-intention consumers is -3.37 percentage points. Thus, the reduction in exploration is not driven solely by the history-dependent construction of the main novelty measure; likely purchase interruption is also associated with a lower propensity to adopt products that have recently entered the market.

We further apply the continuous-score specification in equation (11), replacing the dependent variable with Y_{it}^{NP} . The treatment-by-score coefficient is -4.86 percentage points ($SE = 1.56$; $p = 0.006$) for a one-unit increase in predicted purchase probability. Equivalently, increasing predicted purchase intention by 10 percentage points makes the fitted treated-control post-reopening response in new-product-seeking 0.49 percentage points more negative. The corresponding continuous-score pretrend test does not reject ($p = 0.377$). The binary and continuous specifications therefore lead to the same conclusion: the more likely the closure was to interrupt a purchase, the less consumers subsequently adopt recently appearing products. Appendix D.2 reports the complete coefficient decompositions and diagnostics.

6.3 Effect on Non-Coffee Products

We next examine whether the negative novelty-seeking response is specific to coffee products. Coffee consumption can be strongly habitual, and the caffeine content of many products may further reinforce repeated consumption. If the main result were driven entirely by features specific to coffee, its implications for novelty-seeking in other product categories would be more limited. We therefore re-estimate the same binary DDD specification using only non-coffee products. Because these restrictions substantially reduce the number of qualifying purchasing windows, we view this exercise as a product-scope sensitivity analysis rather than an independent causal test.

Based on the commodity-description fields, we construct three increasingly restrictive product scopes. The broadest definition includes all non-coffee products, including non-coffee beverages, food, and other items. The second retains only non-coffee consumables—beverages and food—and the third retains only

non-coffee beverages. Within each scope, novelty-seeking is defined in the same way as the main outcome: the share of distinct qualifying products purchased in a window that the consumer had not purchased previously. Consequently, the outcome is observed only for windows in which the consumer makes at least one purchase from the corresponding product group.

Across all three product scopes, the estimated DDD remains negative. For all non-coffee products, the estimate is -4.13 percentage points ($SE = 3.40$; restricted wild-cluster $p = 0.225$), remarkably close in magnitude to the -4.15 -point main result. Restricting the outcome to non-coffee consumables produces an estimate of -4.11 percentage points ($SE = 3.39$; restricted wild-cluster $p = 0.230$). Restricting the sample further to non-coffee beverages produces a larger estimate of -6.26 percentage points ($SE = 4.12$; restricted wild-cluster $p = 0.119$). The three point estimates have the same sign as the all-product estimate, although each is imprecisely estimated.

The narrower product definitions substantially reduce the number of purchasing windows on which novelty can be measured: the corresponding estimation samples contain 30,444 observations for all non-coffee products, 30,418 for non-coffee consumables, and 21,433 for non-coffee beverages, compared with 99,644 observations for the main novelty outcome. We therefore use these analyses primarily to examine whether the direction and economic magnitude of the result are specific to coffee, rather than treating them as separate headline estimates.

The accompanying diagnostics also differ across product scopes. For all non-coffee products and non-coffee consumables, the joint high-minus-low pretrend tests reject equality of the leads ($p = 0.006$ in both cases). For non-coffee beverages, the novelty-seeking pretrend test does not reject ($p = 0.226$), while the probability of making a qualifying purchase declines differentially for high-intention treated consumers after reopening: the corresponding DDD is -2.24 percentage points ($SE = 0.90$; $p = 0.023$). Because novelty-seeking is observed only when a qualifying purchase occurs, this extensive-margin response can change the set of consumers entering the beverage-specific novelty regression. We therefore interpret the non-coffee exercise as evidence on the scope of the behavioral pattern: the negative relationship is not unique to coffee products, while the main all-product specification remains the preferred basis for causal inference. Appendix D.3 reports the complete estimates, sample-entry analyses, and event-study diagnostics.

6.4 Accounting for Local Competition

Local competition may affect consumers' product choices after reopening and could therefore contribute to the post-reopening treated-control differences in novelty-seeking. To account for this possibility, we supplement the transaction data with store-address data that report the number of potentially competing establishments located within 500 and 1,500 meters of each store. We use the 500-meter count as our primary measure of local competition and assign it to each consumer through the consumer's preferred store in the pre-closure period. The 1,500-meter count provides an alternative measure using a broader geographic market. Our objective is not to identify local competition as a moderator of the interruption effect, but to assess whether differences in the observed competitive environment can account for the main negative DDD.

Because these competitor counts are time invariant, their levels are absorbed by the consumer fixed effects. We therefore allow local competition to affect changes after reopening by interacting it with the post-period indicator, treatment status, and predicted purchase intention. Let $\tilde{C}_s$ denote the standardized competitor count associated with preferred store s . We augment equation (9) with

$$\text{Post}_{et} \left[\kappa_C \tilde{C}_s + \kappa_{TC} \left(\text{Treated}_{ie} \tilde{C}_s \right) + \kappa_{HC} \left(\text{High}_{ie} \tilde{C}_s \right) + \kappa_{THC} \left(\text{Treated}_{ie} \text{High}_{ie} \tilde{C}_s \right) \right]. \quad (13)$$

The original $\text{Post} \times \text{Treated}$, $\text{Post} \times \text{High}$, and $\text{Post} \times \text{Treated} \times \text{High}$ terms remain in the regression. Because $\tilde{C}_s$ is standardized to have mean zero, the coefficient δ^D on the original triple interaction is the fitted novelty-seeking DDD at the average observed level of local competition. More generally, at a standardized competition level c , the fitted DDD is

$$\delta^D + \kappa_{THC} c.$$

The coefficient κ_{THC} therefore describes how the fitted DDD varies with local competition, while the primary purpose of this specification is to examine whether accounting for the observed competitive environment changes the negative interruption effect.

The main result remains economically similar after introducing these interactions. Using the raw number of potentially competing establishments within 500 meters, the fitted DDD at the mean level of competition is -3.33 percentage points (95% CI: -6.35 to -0.31), compared with -4.15 percentage points in the baseline specification. Using $\log(1 + \text{competitor count})$ instead produces a fitted DDD of -3.92 percentage points (95% CI: -7.64 to -0.20). Expanding the geographic radius to 1,500 meters yields estimates of

−4.04 percentage points using the raw count and −4.11 percentage points using its logarithm. Thus, the negative interruption-related novelty response is similar across alternative measures of the local competitive environment.

We also evaluate the raw 500-meter specification at different levels of competition. At competitor counts of 2, 5, and 13—corresponding to the 25th percentile, median, and 75th percentile among the design stores—the fitted DDDs are −5.03, −4.39, and −2.70 percentage points, respectively. The estimated change in the DDD associated with a one-standard-deviation increase in local competition is +3.00 percentage points, with a 95% confidence interval from −5.15 to 11.14 points. We therefore do not use these estimates to make a separate claim about competition as a moderator; rather, the relevant robustness result is that the fitted interruption effect remains negative over a broad range of observed competitive environments.

Finally, the competitor counts cannot capture every local change that may occur around reopening. We therefore conduct an additional specification that includes preferred-store-by-event-by-post fixed effects. These fixed effects allow each preferred store within each closure event to have its own post-reopening shift and thereby absorb additive local changes—such as neighborhood demand conditions, local promotions, or other store-specific shocks—that are common to high- and low-predicted-intention consumers associated with that store. Identification of the DDD then comes from the differential post-reopening response between high- and low-intention consumers within the same preferred-store and closure-event environment. With these fixed effects, the novelty-seeking DDD is −4.45 percentage points (SE = 1.71; 95% CI: −8.07 to −0.84), again close to the −4.15-point headline estimate.

Taken together, these analyses show that the main novelty-seeking result is not explained by observed differences in local competition or by additive post-reopening shocks shared by consumers associated with the same preferred store. Appendix D.4 reports the construction of the competition measures, alternative geographic definitions, fitted effects at different competition levels, and the corresponding event-study diagnostics.

7 Conclusion

This paper asks how consumers’ willingness to explore changes when a purchase they likely intended to make is unexpectedly blocked. Using temporary store closures and predicted purchase intention, we find that the treated-control post-reopening change in novelty-seeking is 4.15 percentage points lower among

high-intention consumers than among low-intention consumers, equivalent to 8.3% of the pre-period mean. The result is nearly identical when exploration is measured by adoption of products newly introduced by the firm: the estimated DDD is -4.17 percentage points, or 28.6% of that outcome's pre-period mean.

The decomposition of the main DDD is central to the interpretation. Low-intention consumers exhibit a positive closure-related novelty response, consistent with the fact that reduced purchasing during closure leaves more products untried in their observed histories. When the closure is much more likely to interrupt an intended purchase, however, the product-choice response shifts downward by 4.15 points. This high-minus-low difference reveals a behavioral consequence that an average closure effect would obscure: likely interruption pulls subsequent purchasing toward the familiar product set relative to the general product-choice response generated by closure exposure.

The cross-consumer pattern follows the mechanism-linked prediction of familiar-choice salience. Consumers with lower baseline novelty-seeking show the strongest interruption response; across specifications, a one-standard-deviation increase in baseline novelty attenuates the DDD by 2.29–2.94 percentage points. Two concrete exposure-based explanations do not reproduce the pattern. Treated high-intention consumers receive more, rather than fewer, new-product notifications, and their earlier return timing generates only a 0.10-percentage-point reduction in exposure to personally novel assortments, far smaller than the headline effect. Together with the parallel pre-period dynamics and the new-product-seeking result, these findings support the interpretation that an intended but unrealized purchase can carry forward into subsequent product choice.

The managerial implication is that the value of product innovation depends partly on consumers' recent purchase experiences. After store closures, stockouts, app outages, delivery suspensions, or other temporary access constraints, recently interrupted consumers may be less receptive to unfamiliar offerings and more responsive to familiar choices. Firms can use this heterogeneity when timing and targeting new-product launches, designing personalized recommendations, and allocating promotional incentives. When targeting is flexible, familiar products may be more effective recommendations for consumers whose likely purchases were blocked, while early new-product promotion can be directed toward less-affected consumers. When rapid diffusion makes those interrupted consumers important targets, stronger trial incentives such as coupons or price promotions may be needed to offset their greater pull toward familiarity. Purchase interruption therefore shapes not only whether consumers can transact today, but also what they are willing to discover tomorrow.

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Appendix

A Data Construction and Measurement

This appendix records construction details that are needed to interpret the empirical design but are too mechanical for the main text. The unit of assignment is a consumer. A consumer enters the analysis through a closure event for which the consumer's pre-closure purchase history identifies a meaningful focal store; the outcomes then aggregate that consumer's purchases across the observed chain.

A.1 Closure Registry and Consumer Sample

The raw data contain 7,312,498 orders, represented by 10,631,250 product-line records, from 779,744 consumers at 260 stores between June 1, 2020 and December 31, 2021. We aggregate these orders to a store-day panel and define a candidate closure as a consecutive spell in which a store records no transactions. This procedure identifies 101 zero-transaction spells.

We first remove 60 spells at stores located in universities or colleges. These spells occur during summer periods in which COVID-19 policies may have kept students away from campus. Because the data cover purchases made in this city, we do not observe whether students continued to buy from the chain elsewhere. A campus store with no local transactions may therefore reflect the temporary absence of its consumers rather than a supply disruption. Removing these spells avoids treating a seasonal loss of the local consumer population as a store closure and leaves 41 non-campus candidates.

We next exclude seven spells lasting 30 days or longer. The analysis is designed around short disruptions followed by a clearly observed reopening; longer spells are more likely to reflect seasonal shutdowns or persistent changes in store operations. They also produce much longer pre- and post-event windows, which are difficult to compare within the 19-month data period. Requiring closures to last strictly fewer than 30 days leaves 34 candidates.

The remaining events must provide enough support for the treated-control comparison. We require at least one treated and one control consumer and at least 50 consumers in each group. These restrictions remove events for which the DDD is undefined or rests on a very small group. We also require the control purchase rate during the closure to be at least twice the treated rate. This screen is intended to distinguish a localized access disruption from a broader contraction in demand that affects treated and control consumers

alike. Together, the support and purchase-rate screens remove 12 events—eight for a low control purchase rate, two for insufficient group size and two for having no control consumers—leaving 22 closures. Because the purchase-rate screen uses behavior realized during the closure, the full-registry sensitivity analysis remains important.

The maintained headline sample excludes four additional events around the Lunar New Year period. Travel, holiday schedules and COVID-19-related mobility changes during this period can shift consumer traffic independently of a store closure, and these events have weaker treatment-control fit. This restriction leaves 18 closures lasting 10 to 29 days. It is not generated by a coded ex-ante rule, so we retain the 22-event full-registry analysis as a sensitivity check and report it below.

For each retained closure, treated consumers are those whose preferred pre-closure store is the store that closes. Control consumers are those whose preferred store is one of the matched stores that remains open. We process closures chronologically and, without replacement, match each closing store to the five eligible stores with the smallest absolute difference in setup date, a predetermined store characteristic. Candidate controls exclude all treated stores, previously selected controls, and university or college stores. We define the preferred store from pre-closure purchases and require at least five pre-closure purchase days and a preferred-store purchase share of at least 80%. These restrictions ensure that the closing store was an economically meaningful source of access for treated consumers. The final sample contains 40,148 consumers: 7,390 treated and 32,758 controls, each entering only one closure event. Of the treated consumers, 2,041, or 27.6%, have high predicted purchase intention during the closure window.

A.1.1 Sensitivity to the Closure Registry

Table A1 compares the full 22-event registry with the preferred 18-event registry using the same novelty measure and DDD specification. Retaining the four Lunar New Year events produces a DDD of -3.86 percentage points ($SE = 1.67$; $p = 0.031$), close to the -4.15 -point estimate in the preferred registry. The displacement pretrend test does not reject in either sample ($p = 0.707$ and $p = 0.841$, respectively). Thus, the magnitude and inference are not driven by excluding these four events.

Table A1: Novelty-seeking DDD across closure registries

Closure registry	Events	DDD	SE	<i>p</i> -value	Observations	Pretrend <i>p</i> -value
Full registry	22	−3.86	1.67	.031	113,269	.707
Preferred registry	18	−4.15	1.84	.038	99,644	.841

Notes: DDD estimates and standard errors are in percentage points. Both rows use the same member-first novelty outcome, four pre-closure and four post-reopening periods, consumer, relative-period and calendar-month fixed effects, and CRV1 standard errors clustered by closure event. The pretrend column reports the joint test that the three pre-period Treated $\times$ High predicted-intention coefficients equal zero, with period -1 omitted.

A.2 Outcome Timing and Variable Definitions

Each closure event defines relative periods $t = -4, \dots, 4$. Every relative period has the same length as the closure, denoted L_e , and period 0 is the closure window itself. The main post-reopening regressions omit period 0 and compare four pre-closure windows with four post-reopening windows. This convention produces 321,184 consumer-period observations for purchase frequency. Consumers make at least one purchase in 105,854 of these rows, so novelty-seeking is defined in those rows. The consumer fixed effects then remove 6,210 rows belonging to consumers with only one product-choice period, leaving 99,644 observations in each main product-choice regression. Table 2 reports pre-period purchase probabilities across the four DDD cells. Table A2 summarizes the construction and interpretation of the variables used throughout the empirical analysis.

Table A2 highlights two distinctions that govern the interpretation of the results. First, treatment indicates exposure to a focal-store closure, whereas a likely interrupted purchase requires both treatment and high predicted purchase intention. Second, purchase frequency is defined for every consumer-period, while novelty-seeking and new-product-seeking describe product composition only in windows containing a purchase. Baseline novelty-seeking is measured before the focal closure and is used only in the heterogeneity analysis. These distinctions separate the extensive-margin response from the conditional product-choice response and motivate the sample-entry analysis below.

A.3 Purchase Probability and the Novelty-Seeking Sample

Novelty-seeking is defined only in consumer windows containing a purchase. Figure 4 in the main text plots the realized probability of at least one purchase in each DDD cell and relative period. This is also the probability that a window enters the novelty-seeking sample. Purchase probability is much higher among

Table A2: Key variables in the empirical design

Variable	Construction	Interpretation
Treated	Preferred pre-closure store is the closed store.	Exposure to the local store closure. This is not the same as having a purchase blocked.
Control	Preferred pre-closure store is a matched store that remains open.	Counterfactual local-store path for treated consumers.
High predicted purchase intention	Ex-ante predicted probability of at least one purchase during the closure-length window is at least .5.	Predicted closure-window purchase intention. It is defined before observing post-reopening outcomes.
Interrupted purchase	Treated $\times$ High predicted purchase intention.	The closure is predicted to bind: the consumer is exposed to the closure and likely would have purchased during the closure window.
Post	Relative periods 1 through 4.	Post-reopening windows. Period 0 is the closure window and is omitted from the main DDD regressions.
Purchase frequency	Days with at least one purchase at any chain store in the relative-period window, divided by L_e .	Chain-wide extensive-margin purchase intensity, normalized for closure length; multiple orders on one day count once.
Novelty-seeking	Share of distinct products purchased in the window that the consumer had not previously purchased.	Chain-wide product exploration relative to the consumer's own observed history, conditional on purchase.
Baseline novelty-seeking	Consumer-level mean novelty-seeking over relative periods -8 through -1 , restricted to consumers with at least five orders over those periods and standardized over eligible consumers.	Revealed baseline orientation toward familiar versus novel products. Relative periods are closure-length windows, not calendar weeks.
New-product-seeking	Share of distinct products purchased whose first observed chain sale occurs in the current or immediately previous event window.	Adoption of recently appearing products, conditional on purchase. First observed sale need not equal the formal catalog launch date.

Notes: The main text uses “high predicted purchase intention” for consumers with a score above the threshold. A likely interrupted purchase occurs only for treated high-predicted-intention consumers, because controls retain access to their focal store.

high-predicted-intention consumers, as expected from a classifier designed to predict purchasing. The informative comparison is treated versus control within each intention stratum, where the paths are close. High-intention purchase probability falls from 0.66–0.85 across the pre-periods to roughly 0.56–0.61 after reopening, whereas low-intention purchase probability remains near 0.17–0.27. The conditional novelty regressions thus compare purchasing windows drawn from a population whose composition changes over time. Similar treated and control purchase-probability paths reduce concern about differential sample entry, but they do not establish that the observed purchasers are compositionally identical. The main estimates therefore retain a conditional-on-purchase interpretation.

A.4 Predicted Purchase-intention Classifier

The classifier separates closure exposure from likely purchase interruption. It predicts whether a consumer would purchase at least once in a window with the same length as the closure. This target depends mechanically on window length, so the implementation trains one XGBoost model for each observed closure duration. Training uses pre-closure relative periods -4 through -2 , validation uses the holdout pre-period -1 , and an additional audit uses control consumers in period 0 because their matched focal stores remain open. The period-0 audit is a validation exercise; the DDD uses the ex-ante predicted score. Table A3 reports the prediction target, information set, validation samples, and classification rule.

Table A3: Classifier implementation details

Component	Implementation
Prediction target	Indicator for at least one purchase in a closure-length window.
Model family	XGBoost binary classifier, estimated separately by closure duration.
Training windows	Relative periods -4 , -3 and -2 .
Holdout validation	Relative period -1 , separately audited for treated and control consumers.
Additional audit	Control consumers in period 0, where the matched control store remains open and actual purchase behavior is observed.
Scored sample	Treated and control consumers in the ex-ante period-0 panel.
Classification rule	High predicted purchase intention equals one if the predicted period-0 purchase probability is at least .5.
Feature families	Recent purchase frequency, recency, total purchase history, preferred-store loyalty, basket composition, discount and coupon use, day-of-week purchasing patterns, demographics and push-notification history.

Notes: The performance table in the main text and the diagnostics below use the exact final 18-event cohort. The implementation table emphasizes timing and information sets, because these are the key design concerns for interpreting the classifier as ex ante. The .5 cutoff is fixed in the classifier configuration and is not selected using the post-reopening outcomes.

The implementation in Table A3 keeps the score ex ante. Every feature is constructed from information available before the prediction window, and the period-0 outcomes for control consumers are used only to evaluate predictions after the model has been trained. The .5 cutoff is likewise fixed before estimating the post-reopening outcomes. The classifier therefore does not use the outcomes that enter the DDD to determine either the score or the high-intention classification.

Table A4 reports classification performance at the .5 cutoff in the period-0 control audit and the two period- -1 holdout samples.

The classifier improves on the no-purchase benchmark in each evaluation sample, although the improvement is modest in the period-0 control audit. In that audit, accuracy is 0.756 compared with a benchmark of

Table A4: Purchase-intention classification at the .5 cutoff

Evaluation sample	TN	FP	FN	TP	Accuracy	Benchmark	Sensitivity	Specificity	PPV	NPV	F1
Control, $t = 0$	20,045	4,111	3,889	4,713	.756	.737	.548	.830	.534	.838	.541
Control, $t = -1$	19,456	3,214	4,321	5,767	.770	.692	.572	.858	.642	.818	.605
Treated, $t = -1$	4,368	698	971	1,353	.774	.686	.582	.862	.660	.818	.619

Notes: $t = 0$ is the closure window and $t = -1$ is the holdout pre-period. TN, FP, FN and TP denote true negatives, false positives, false negatives and true positives. Benchmark is the accuracy of predicting no purchase for every consumer. Sensitivity is $TP/(TP+FN)$, specificity is $TN/(TN+FP)$, PPV is $TP/(TP+FP)$, and NPV is $TN/(TN+FN)$. All rows use the exact final cohort.

0.737. Specificity is 0.830, while sensitivity is 0.548, indicating that the .5 cutoff identifies nonpurchasers more accurately than purchasers. Performance is somewhat stronger in the period- -1 holdouts: accuracy is 0.770 for controls and 0.774 for treated consumers, with positive predictive values of 0.642 and 0.660. Thus, the binary rule provides meaningful separation between consumers with different purchase propensities, but it does not classify every realized purchaser correctly.

Figure A1 examines whether the predicted probabilities also match observed purchase rates.

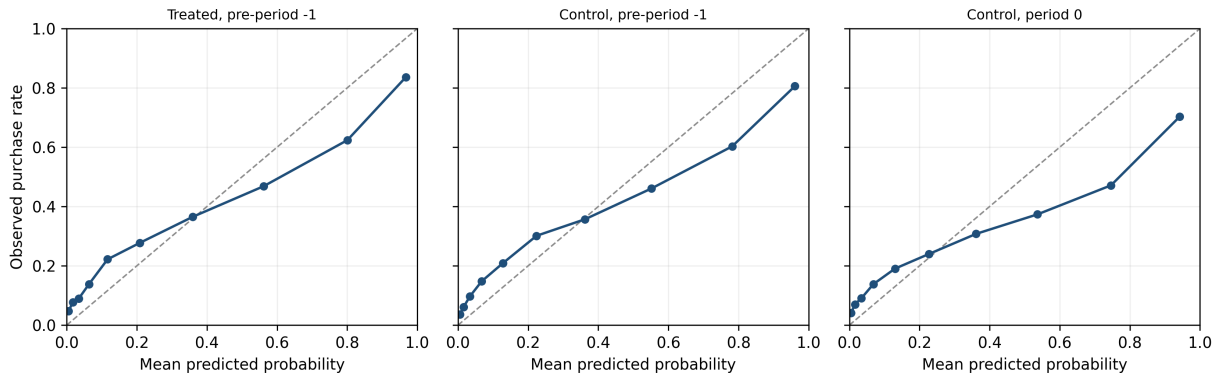


Figure A1: Calibration of predicted purchase intention

Notes: Each point is one decile of predicted purchase probability in the exact final cohort. The horizontal coordinate is the decile's mean prediction and the vertical coordinate is its observed purchase rate. The dashed 45-degree line denotes perfect calibration. The flatter empirical paths show that the raw probabilities are too dispersed, although they continue to rank purchase intention.

The calibration paths in Figure A1 are flatter than the 45-degree line in all three evaluation samples. The model therefore spreads predicted probabilities too widely: observed purchase rates vary less across score deciles than the raw predictions imply. At the same time, observed purchase rates rise with the predicted score, consistent with the ROC-AUC and PR-AUC evidence in the main text that the model meaningfully ranks consumers by purchase propensity. The score is therefore more reliable as a relative ranking than as a perfectly calibrated probability.

Table A5 evaluates whether this predictive performance is concentrated in closure events of particular

lengths.

Table A5: Period-0 control performance by closure duration

Duration	<i>N</i>	Prevalence	ROC-AUC	PR-AUC	Brier	Calibration slope
10 days	1,244	.210	.774	.501	.157	.461
11 days	5,020	.246	.783	.585	.152	.548
12 days	3,164	.196	.786	.508	.154	.432
13 days	1,662	.238	.780	.525	.179	.388
14 days	5,232	.230	.783	.555	.148	.592
15 days	551	.376	.754	.686	.210	.374
16 days	2,416	.300	.794	.646	.176	.448
17 days	2,783	.304	.789	.627	.183	.478
18 days	3,176	.224	.786	.569	.155	.537
26 days	4,645	.299	.761	.592	.208	.494
28 days	1,891	.347	.791	.689	.199	.489
29 days	974	.359	.739	.611	.232	.331

Notes: This table evaluates control consumers in period 0 separately for each duration-specific classifier. Duration 25 does not occur in the final 18-event registry and is therefore excluded; it appears only in a broader sensitivity sample. Definitions follow Table 1.

Predictive ranking remains informative across all closure durations in Table A5. Period-0 control ROC-AUC ranges from 0.739 for the 29-day model to 0.794 for the 16-day model, and PR-AUC exceeds the corresponding purchase prevalence in every row. Calibration is consistently weaker: the duration-specific slopes range from 0.331 to 0.592, all below the value of one associated with perfect calibration. No single closure duration therefore drives the classifier’s overall ranking performance or its tendency to produce overly dispersed probabilities.

Table A6 shows how the period-0 control classification changes when the cutoff varies from .3 to .7.

Table A6: Predictive tradeoffs across classification cutoffs

Cutoff	Predicted high	Accuracy	Sensitivity	Specificity	PPV
.30	.391	.712	.695	.718	.468
.40	.325	.738	.620	.780	.502
.50	.269	.756	.548	.830	.534
.60	.216	.770	.474	.875	.575
.70	.170	.777	.398	.911	.615

Notes: The evaluation sample is the exact-cohort period-0 controls. Predicted high is the share classified above the stated probability cutoff. These are predictive tradeoffs, not DDD estimates. The main .5 rule is configured before estimation; the continuous-score DDD avoids discretizing the score.

The cutoff exercise displays the expected sensitivity-specificity tradeoff. Lowering the cutoff from .5 to .3 increases the share classified as high intention from 26.9% to 39.1% and raises sensitivity from 0.548

to 0.695, but reduces specificity and positive predictive value. Raising the cutoff to .7 reduces the high-intention share to 17.0% and increases specificity and positive predictive value to 0.911 and 0.615, while sensitivity falls to 0.398. The .5 rule provides an intermediate classification rather than a uniquely optimal cutoff, which motivates the continuous-score robustness specification.

Figure A2 compares the distributions of the predicted period-0 score between treated and control consumers.

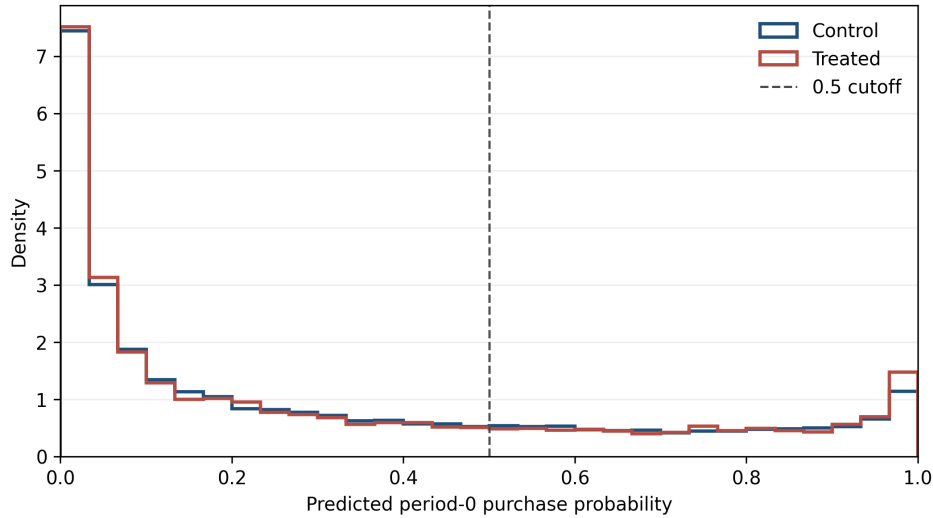


Figure A2: Overlap in predicted period-0 purchase intention

Notes: The figure shows density histograms of the ex-ante score in the exact final cohort. Mean predicted intention is 0.307 for controls and 0.313 for treated consumers; the median is 0.172 and 0.174, respectively. The shares at or above 0.5 are 26.9% and 27.6%. The similar distributions indicate strong score overlap across treatment arms, although overlap does not validate calibration for the unobserved treated period-0 counterfactual.

The treated and control score distributions in Figure A2 are closely aligned. Their means are 0.313 and 0.307, their medians are 0.174 and 0.172, and 27.6% and 26.9% of observations exceed the .5 cutoff, respectively. This overlap indicates that the DDD does not compare treatment groups occupying disjoint regions of the predicted-score distribution. Taken together, the diagnostics show that the classifier ranks purchase propensity reasonably well across closure durations and treatment groups, while calibration is consistently weaker and no cutoff is uniquely preferred. They therefore motivate cautious language for both the binary classification and the continuous-score robustness specification; they do not turn predicted purchase propensity into a direct measure of psychological intention.

B Identification Diagnostics

This appendix reports the formal diagnostics behind the identification discussion in Section 4.3. The main design has three dimensions: treatment exposure, post-reopening timing and predicted purchase intention. The key identifying restriction is therefore a triple-difference restriction: absent the closure, the treated-control gap for high-predicted-intention consumers would have changed like the corresponding gap for low-predicted-intention consumers.

B.1 Pretrend Tests

For the main novelty-seeking result, we estimate the event-study analog of equation (9):

$$Y_{iet}^N = \alpha_{ie} + \lambda_t + \gamma_m + \sum_{\ell \in \{-4, -3, -2, 1, 2, 3, 4\}} \left[\delta_\ell^B \text{Treated}_{ie} + \beta_\ell \text{High}_{ie} + \delta_\ell^D (\text{Treated}_{ie} \times \text{High}_{ie}) \right] \mathbb{I}\{t = \ell\} + \varepsilon_{iet}. \quad (\text{A1})$$

Period -1 is the omitted reference period, and period 0 is the closure window and is excluded. The coefficient δ_ℓ^D is the high-minus-low DDD at relative period ℓ . The pretrend test is the joint null $H_0 : \delta_{-4}^D = \delta_{-3}^D = \delta_{-2}^D = 0$.

Table A7 reports the three lead coefficients and their joint test for the main novelty-seeking outcome.

Table A7: Novelty-seeking DDD pretrend coefficients and joint test

<i>Panel A: Pre-period high-minus-low DDD coefficients</i>			
Relative period	DDD estimate	SE	95% CI
−4	2.23	2.73	[−3.52, 7.98]
−3	2.64	3.44	[−4.61, 9.90]
−2	1.60	1.98	[−2.57, 5.77]
<i>Panel B: Joint test</i>			
Null hypothesis	Wald statistic	Restrictions	p-value
$\delta_{-4}^D = \delta_{-3}^D = \delta_{-2}^D = 0$	.837	3	.8407

Notes: Coefficients and standard errors are in percentage points. The dependent variable is novelty-seeking, defined only in windows with a purchase. The regression uses 99,644 observations and includes consumer, relative-period and calendar-month fixed effects. Period -1 is the omitted reference, and period 0 is excluded. Standard errors are clustered by closure event (18 clusters). The joint Wald test uses the closure-clustered covariance matrix.

The three novelty-seeking lead estimates in Table A7 are positive, between 1.60 and 2.64 percentage points, and each confidence interval includes zero. The joint test does not reject that all three are zero.

This is evidence against a pronounced differential pretrend in the observed pre-closure windows, but it does not prove the identifying assumption. Replacing the binary high-intention indicator with the continuous predicted-intention score likewise does not reject a differential pretrend (Wald statistic = 1.781, $p = 0.619$). For transparency about the other outcomes, the purchase-frequency high-minus-low test rejects ($p = 0.035$), which is why that result is treated as descriptive, while the new-product-seeking robustness outcome does not reject ($p = 0.114$).

Figure A3 shows the novelty high-minus-low coefficient path. The omitted reference period is -1 , and period 0 is the closure window and is excluded from estimation. Each displayed 95% confidence interval includes zero. The post-reopening point estimates are negative, but individually imprecise; the figure is therefore evidence about the timing and uncertainty of the response, not a claim that every post-period coefficient differs from zero.

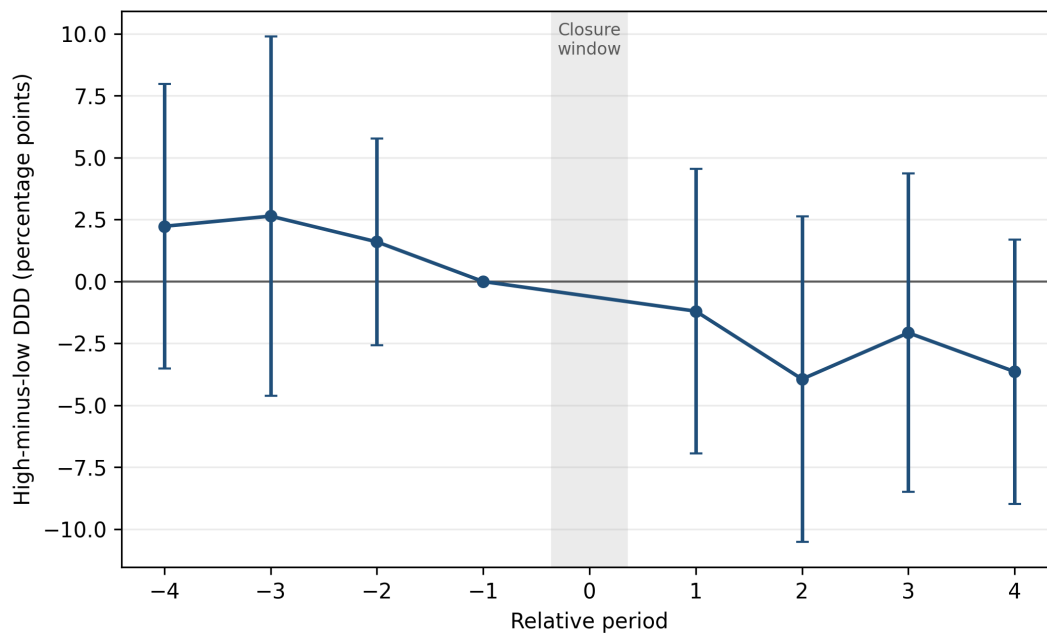


Figure A3: Event-study path of the novelty-seeking DDD

Notes: Points are the event-study coefficients on Treated $\times$ High predicted intention relative to period -1 . Vertical bars use the confidence limits reported by the closure-event-clustered estimator (18 clusters). Period 0 is the closure window and is omitted. Novelty-seeking is conditional on a purchase in the window.

B.2 Dynamics of Baseline-Novelty Heterogeneity

The collapsed heterogeneity estimates in Table 4 compare all four post-reopening periods with all four pre-closure periods. Figure A4 reports the dynamic counterpart to the order-history-adjusted specification in column (2), replacing the collapsed post indicator with relative-period interactions while retaining the same fixed effects, baseline-order adjustment and closure-event clustering. The omitted reference period is -1 .

The three lead coefficients are small relative to their confidence intervals, and their joint test does not reject equality to zero ($p = 0.561$). The post-period gradients are positive in periods 1, 3 and 4 and near zero in period 2. The largest estimate occurs immediately after reopening, but every period-specific 95% confidence interval includes zero. The figure therefore supports the absence of a pronounced differential pretrend while also showing that the timing of the mechanism result is imprecisely estimated.

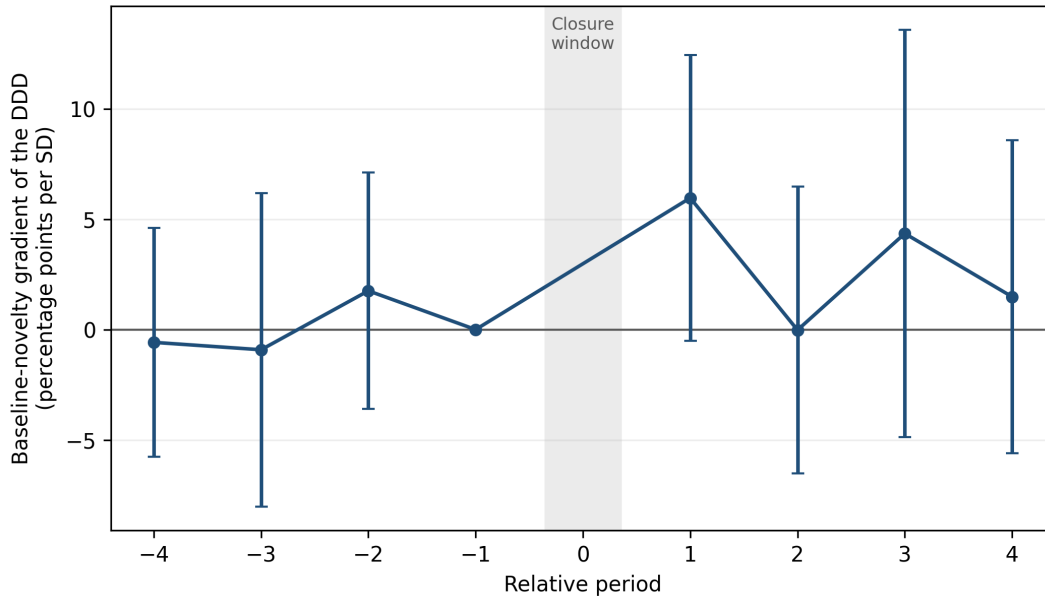


Figure A4: Event-study path of the baseline-novelty gradient

Notes: Points are coefficients on Relative period $\times$ Treated $\times$ High predicted intention $\times$ standardized baseline novelty-seeking. Vertical bars are 95% confidence intervals from CRV1 standard errors clustered by closure event (18 clusters). Relative period -1 is the omitted reference period; period 0 is the closure window and is excluded. The specification includes the full lower-order interaction hierarchy for baseline novelty and the parallel hierarchy for standardized log baseline orders, corresponding to column (2) of Table 4.

C Alternative Explanations

C.1 New-Product Notification Exposure

This section reports the construction and estimates behind Section 5.2.1. The raw files contain 56,724,521 push records. We retain the 40,148 consumers in the 18-closure analysis and map notifications into the four event-length periods before closure, the closure window and the four event-length periods after reopening. This produces a balanced panel of 361,332 consumer-period observations. Of the 912,911 push records mapped into these windows, 125,918 have `trigger_tag=3`, the platform classification for a new-product notification.

The primary measure counts unique consumer-policy-date notification campaigns and divides by the number of calendar days in the event window. All 125,918 new-product records are unique at this grain, so the campaign and raw-record measures coincide. Supporting outcomes indicate whether a consumer receives any new-product notification, count distinct notification days and measure the share of all campaigns classified as new-product notifications. The source files record targeting entries; they do not verify delivery, impression, opening or reading.

For exposure outcome N_{iet} , we estimate

$$N_{iet} = \beta_1(\text{After}_{et} \times \text{Treated}_i) + \beta_2(\text{After}_{et} \times \text{High}_i) + \beta_3(\text{After}_{et} \times \text{Treated}_i \times \text{High}_i) + \phi_{ie} + \omega_t + \gamma_m + \varepsilon_{iet}, \quad (\text{A2})$$

where After_{et} denotes either the closure window or the post-reopening periods. The closure specification compares period 0 with all four pre-periods; a sensitivity check compares it only with period -1 . The post specification omits period 0 and compares the four post-periods with the four pre-periods, as in the main outcome regression. Standard errors are clustered by closure event. Because there are 18 clusters, Table A8 also reports restricted wild-cluster p -values for β_3 .

Table A8 shows that the differential change in recorded new-product campaigns is positive rather than negative. Relative exposure among treated high-intention consumers increases by 0.0041 campaigns per day during the closure when all pre-periods form the comparison ($p = 0.003$) and by 0.0043 when period -1 alone forms the comparison ($p = 0.011$). The differential remains positive after reopening at 0.0017 campaigns per day ($p = 0.047$). The indicators for receiving any new-product notification and the number of notification days produce the same positive pattern. Total campaign frequency also rises, although its

Table A8: Differential exposure to new-product notifications

	During versus all pre-periods (1)	During versus period -1 (2)	Post versus pre-periods (3)
New-product campaigns per day	0.0041 (0.0011)	0.0043 (0.0013)	0.0017 (0.0007)
Restricted wild-cluster p -value	[0.0032]	[0.0111]	[0.0469]
Any new-product notification	0.0422 (0.0100)	0.0362 (0.0110)	0.0144 (0.0061)
Restricted wild-cluster p -value	[0.0010]	[0.0057]	[0.0399]
New-product notification days per day	0.0041 (0.0011)	0.0043 (0.0013)	0.0017 (0.0007)
Restricted wild-cluster p -value	[0.0031]	[0.0111]	[0.0471]
All campaigns per day	0.0230 (0.0087)	0.0143 (0.0078)	0.0113 (0.0058)
Restricted wild-cluster p -value	[0.0823]	[0.1648]	[0.1104]
New-product share of campaigns	0.0020 (0.0095)	−0.0014 (0.0057)	−0.0005 (0.0058)
Restricted wild-cluster p -value	[0.8505]	[0.8080]	[0.9406]
Observations, primary outcome	200,740	80,296	321,184
Closure-event clusters	18	18	18

Notes: Entries are coefficients on $\text{After} \times \text{Treated} \times \text{High}$ from equation (A2). A negative coefficient is required for the proposed notification explanation. Standard errors clustered by closure event are in parentheses; restricted wild-cluster p -values are in brackets. All specifications include consumer, relative-period and calendar-month fixed effects. The campaign-share regressions are conditional on receiving at least one campaign and use 107,606, 39,628 and 183,480 observations in columns (1)–(3), respectively.

wild-cluster p -values exceed 0.05, while the new-product share of all campaigns remains close to zero in every comparison. The increase in new-product notifications therefore reflects a broader increase in recorded campaign targeting rather than a larger new-product share, and none of the specifications produces the negative differential required by the notification explanation.

Table A9 reports the dynamic specification, which replaces *After* with relative-period indicators and omits period -1 . The three lead estimates are individually small, and their joint test does not reject equality to zero ($p = 0.500$). The coefficient is largest during closure, remains positive in the first reopening period and then declines. This timing is consistent with high-intention treated consumers becoming newly eligible for inactivity-triggered campaigns when a likely purchase is interrupted. It is inconsistent with the negative differential exposure needed to explain the novelty DDD.

The inactivity pattern in the records also clarifies why absolute high-low levels are not informative. Among treated consumers, the median new-product notification occurs 121 days after the last purchase for low-intention consumers before closure and 129 days during closure. The corresponding medians for high-intention consumers are 56 and 40 days. Low-intention consumers therefore receive more notifications in

Table A9: Event-study path of new-product notification exposure

Relative period	Coefficient	Standard error	<i>p</i> -value
−4	0.0007	(0.0010)	[0.466]
−3	−0.0002	(0.0012)	[0.878]
−2	0.0002	(0.0009)	[0.800]
0	0.0043	(0.0013)	[0.004]
1	0.0031	(0.0013)	[0.036]
2	0.0015	(0.0011)	[0.185]
3	0.0022	(0.0011)	[0.064]
4	0.0003	(0.0008)	[0.712]
Joint pretrend <i>p</i> -value		0.500	

Notes: The outcome is unique new-product notification campaigns per consumer-day. Coefficients are relative-period $\times$ Treated $\times$ High interactions relative to period -1 . Standard errors use CRV1 clustering by the 18 closure events. Period 0 is the closure window.

levels because they are ordinarily more dormant. The DDD removes this persistent gap and shows that the relative exposure increase occurs among treated high-intention consumers instead.

C.2 Reopening Assortment

This section reports the construction and estimates behind the assortment test in Section 5.2.2. The alternative explanation is that high- and low-intention consumers may return at different times after a treated store reopens and therefore encounter different product assortments. For differences in return timing to generate differences in product choice, however, the assortment available at the store must itself vary systematically with time since reopening. We therefore focus first on the evolution of realized assortment at the 18 treated stores during the first four weeks after reopening.

For each closure, we construct four fixed seven-day periods beginning immediately after reopening. Fixed weeks provide a common measure of time since reopening across events even though the preceding closures last between 10 and 29 days. We construct three measures from product-level transactions. *Core-product coverage* defines a store’s core set as products sold in at least three of the four weeks before closure and measures the share of that set sold in each post-reopening week. *Products per 50 baskets* draws 50 consumer-day baskets without replacement, counts the number of distinct products, and averages this count over 100 draws. This rarefaction holds observed basket volume fixed; store-weeks with fewer than 50 baskets are missing. *Pre-menu overlap* is the Jaccard similarity between products sold in a store-week and the union of products sold in that store’s four pre-closure weeks. These measures capture different dimensions

of the realized product assortment while avoiding a mechanical interpretation of lower transaction volume as lower product availability.

Assortment evolution after reopening. For assortment outcome A_{et} at treated store associated with closure event e in post-reopening week t , we estimate

$$A_{et} = \alpha_e + \sum_{k=2}^4 \beta_k \mathbb{I}\{t = k\} + \varepsilon_{et}, \quad (\text{A3})$$

where week 1 is the omitted category and α_e denotes closure-event fixed effects. The coefficients β_k therefore measure how realized assortment in weeks 2–4 differs from the first week after reopening. We test the joint null

$$H_0 : \beta_2 = \beta_3 = \beta_4 = 0,$$

which corresponds to no systematic change in the assortment across the four post-reopening weeks. Table A10 reports the weekly coefficients and joint tests for all three measures.

Table A10: Evolution of realized assortment after reopening

	Core-product coverage (1)	Products per 50 baskets (2)	Pre-menu overlap (3)
Week 2 relative to week 1	0.0059 (0.0185)	0.3019 (0.7351)	−0.0100 (0.0129)
Week 3 relative to week 1	−0.0648 (0.0544)	−0.7031 (0.6006)	−0.0763 (0.0410)
Week 4 relative to week 1	−0.0840 (0.0537)	−1.6600 (0.5552)	−0.1111 (0.0403)
Joint equality test, p -value	0.3821	0.0106	0.0175
Week 1 mean	0.7687	29.2581	0.6120
Observations	72	64	72
Treated stores	18	16	18

Notes: The sample contains the treated stores during the first four fixed seven-day periods after reopening. Core-product coverage is the share of products in the store’s core pre-closure set that sell in the current week. Products per 50 baskets averages the number of distinct products across 100 samples of 50 consumer-day baskets drawn without replacement. Pre-menu overlap is the Jaccard similarity between products sold in the current week and the store’s pre-closure product set. Week 1 is the omitted category in equation (A3). Standard errors in parentheses use CRV1 clustering by closure event. The joint row reports restricted Rademacher wild-cluster p -values based on 9,999 replications for the null that the coefficients for weeks 2, 3, and 4 are jointly zero. Columns (1) and (3) contain all 18 treated stores; column (2) requires at least 50 baskets in every week and therefore contains 16.

Table A10 shows no statistically detectable change in core-product coverage: relative to week 1, the estimates are +0.59, −6.48, and −8.40 percentage points in weeks 2–4, and the joint test yields $p = 0.382$.

Standardized product breadth first increases by 0.30 products per 50 baskets and then falls by 0.70 and 1.66 products, with a joint p -value of 0.011. Pre-menu overlap similarly declines by 1.00, 7.63, and 11.11 percentage points, and its joint test rejects a constant path ($p = 0.018$). Assortment stability alone therefore cannot rule out differential return timing.

Return timing and consumer-specific novel opportunity. We next test the channel directly among treated consumers. A returner is a treated consumer whose first purchase at the reopened focal store occurs within 28 days of reopening. Of 5,349 low-intention and 2,041 high-intention treated consumers, 1,134 (21.2%) and 1,268 (62.1%), respectively, return within this window. High-intention returners return 2.87 days earlier after conditioning on closure-event fixed effects ($SE = 0.29$; $p < 0.001$). Their median return is day 7 rather than day 11, and 53.6% return in week 1 compared with 36.9% of low-intention returners.

For each returner, we construct the realized product set in each of the four candidate return weeks. To prevent the consumer's observed purchase from mechanically entering the opportunity set, we remove any product sold only to that consumer in the store-week; products also purchased by another consumer remain. We define a product as personally novel if the consumer had not purchased it by the reopening date. Let M_{iew} denote either the leave-one-consumer-out novel-product share or count for consumer i , event e , and candidate return week w . Our timing measure is

$$D_{ie} = M_{ie,w(i)} - \frac{1}{4} \sum_{w=1}^4 M_{iew}, \quad (\text{A4})$$

where $w(i)$ is the consumer's actual return week. Subtracting the same consumer's four-week mean removes persistent high-low differences in product history and retains only the opportunity change associated with returning in one week rather than another. We regress D_{ie} on the high-intention indicator and closure-event fixed effects, clustering by closure event. Table A11 reports the return-time difference together with the resulting differences in personally novel product opportunity.

Table A11: Return timing and personally novel product opportunity

	Days to first return	Novel-product share	Novel-product count
	(1)	(2)	(3)
High minus low intention	−2.8724	−0.0010	−0.2756
	(0.2936)	(0.0006)	(0.3848)
	[< 0.001]	[0.0930]	[0.4836]
Low-intention mean	12.1570	−0.0005	1.5780
High-intention mean	9.0331	−0.0016	1.0562
Observations	2,402	2,325	2,402
Closure-event clusters	18	17	18

Notes: The sample contains treated consumers whose first focal-store purchase occurs within 28 days after reopening. Column (1) reports days from reopening to that purchase. Columns (2) and (3) report the timing deviations in equation (A4); means in those columns are therefore deviations from each consumer’s four-week average opportunity, not levels. The share is the fraction of the leave-one-consumer-out realized product set that the consumer had never purchased by reopening, and the count is the number of such products. The share analysis requires a nonempty leave-one-consumer-out product set in all four weeks. All specifications include closure-event fixed effects. Standard errors clustered by closure event are in parentheses, and two-sided p -values are in brackets.

Table A11 confirms that high-intention returners return 2.87 days earlier on average. Their mean return time is 9.03 days, compared with 12.16 days for low-intention returners. Despite this timing difference, the high-minus-low deviation is only −0.10 percentage points for the novel-product share (95% CI: −0.22 to +0.02) and −0.28 products for the count (95% CI: −1.09 to +0.54). The count estimate is statistically indistinguishable from zero and small relative to roughly 67 personally novel products in the realized return-week sets. The return-timing analysis shows a small measured difference in novel-product opportunity between high- and low-intention consumers returning to the focal store.

Comparison with matched control stores. As an additional check, we examine whether treated stores experience an unusual change in realized assortment relative to their matched control stores. This analysis uses the 18 treated stores and their five matched control stores and constructs four fixed seven-day periods before closure and four fixed seven-day periods after reopening. For assortment outcome A_{est} at store s , event e , and relative week t , we estimate

$$A_{est} = \beta(\text{TreatedStore}_{es} \times \text{Post}_{et}) + \alpha_{es} + \lambda_{et} + \epsilon_{est}, \quad (\text{A5})$$

where α_{es} denotes event-store fixed effects and λ_{et} denotes event-by-relative-week fixed effects. The specification compares the change in assortment at the treated store with the contemporaneous change among its matched controls.

Table A12 reports no evidence of a relative contraction after reopening. Core-product coverage falls by 1.29 percentage points more at treated stores than at controls (SE = 3.41 percentage points; $p = 0.710$). Standardized product breadth increases by 0.31 products per 50 baskets (SE = 0.52; $p = 0.562$), while pre-menu overlap increases by 1.46 percentage points (SE = 2.50 percentage points; $p = 0.567$). The immediate-reopening estimates are also not statistically distinguishable from zero for any measure. The joint pretrend tests yield p -values of 0.110, 0.412, and 0.522 for core-product coverage, product breadth, and pre-menu overlap, respectively. Thus, neither the collapsed estimates nor the event-study comparisons show a common relative post-reopening decline at treated stores.

Table A12: Realized assortment at reopening stores and matched controls

	Core-product coverage (1)	Products per 50 baskets (2)	Pre-menu overlap (3)
Post $\times$ Treated store	−0.0129 (0.0341) [0.7097]	0.3063 (0.5178) [0.5619]	0.0146 (0.0250) [0.5665]
Immediate reopening, relative to week −1	0.0205 (0.0365) [0.5806]	0.1935 (0.8753) [0.8277]	0.0495 (0.0293) [0.1097]
Joint pretrend p -value	0.1103	0.4123	0.5218
Control pre-period mean	0.9240	28.2045	0.7513
Observations	736	723	736
Event-store pairs	92	92	92
Closure-event clusters	18	18	18

Notes: Core-product coverage is the share of products in a store's core pre-closure set that sell in the current week. Products per 50 baskets averages the number of distinct products across 100 samples of 50 consumer-day baskets drawn without replacement.

Pre-menu overlap is the Jaccard similarity between products sold in the current week and the store's pre-closure product set.

Standard errors clustered by the 18 closure events are in parentheses, and two-sided p -values are in brackets. Product sales measure realized assortment and need not equal the product set displayed or available to every consumer.

The matched-store analysis addresses a different question from the return-timing test. It asks whether closure and reopening are associated with an unusual store-level assortment change relative to contemporaneous controls. The four-week path shows that realized assortment subsequently evolves, so the direct test of the alternative is instead the consumer-specific timing deviation in equation (A4). The transaction data cannot reveal the exact product set presented to each consumer, and the return analysis is descriptive among

consumers who return within 28 days. Subject to these limitations, the observed timing-induced differences are too small to account quantitatively for the main novelty result.

D Additional Results

D.1 Complete Main Specifications

The main text reports the group-specific effects and DDDs that carry the economic interpretation. This section reports the full coefficient vectors from the underlying regressions. Table A13 presents equation (9) for the two main outcomes.

Table A13: Complete binary predicted-intention specifications

	Purchase frequency (1)	Novelty-seeking (2)
Post $\times$ Treated (δ^B)	.0009 (.0024) [.7107]	.0313 (.0178) [.0976]
Post $\times$ High predicted intention (β)	-.0344 (.0030) [<.001]	.0381 (.0081) [<.001]
Post $\times$ Treated $\times$ High predicted intention (δ^D)	.0058 (.0032) [.0860]	-.0415 (.0184) [.0380]
Pre-period outcome mean	.0493	.4978
Observations	321,184	99,644
R^2	.562	.416
Within R^2	.013	.001
Consumer fixed effects	Yes	Yes
Relative-period fixed effects	Yes	Yes
Calendar-month fixed effects	Yes	Yes

Notes: This table reports every coefficient from equation (9). Post $\times$ Treated is the low-predicted-intention treated-control post contrast. Post $\times$ High captures the high-minus-low post shift common to treated and control consumers and is not a treatment effect. The triple interaction is the high-minus-low DDD and is shown in boldface. The closure window is omitted.

Novelty-seeking is conditional on a purchase in the window. Standard errors, clustered by closure event (18 clusters), are in parentheses; p-values are in brackets. Table 3 reports the novelty-seeking decomposition highlighted in the main text.

Table A13 reports the complete binary specifications for novelty-seeking and the auxiliary purchase-frequency outcome, including the low-group contrasts, focal DDDs, and common high-minus-low post shifts. For purchase frequency, the low-group contrast is 0.0009 and the DDD is 0.0058, while the common post shift is -0.0344 ($p < 0.001$). For novelty-seeking, the corresponding coefficients are 0.0313, -0.0415 , and 0.0381 ($p < 0.001$ for the common shift). Thus, high-predicted-intention consumers have lower post-period purchase frequency and higher post-period novelty-seeking than low-predicted-intention consumers after pooling treated and control observations. These common shifts reflect the relation between predicted purchase timing and post-period behavior; they are not closure effects. The DDD removes this common

high-minus-low movement and isolates the additional treated high-group response under the identifying assumptions. The algebraically equivalent group-effect parameterization in Table 3 then combines δ^B and δ^D with covariance-aware standard errors.

D.1.1 Continuous predicted-intention score

To avoid relying on the 0.5 classification threshold, we replace High_{ie} with the mean-centered predicted purchase-intention score $\tilde{s}_{ie}$:

$$Y_{iet} = \delta_0^B \cdot \text{Post} \times \text{Treated} + \beta^S \cdot \text{Post} \times \tilde{s}_{ie} + \delta^S \cdot \text{Post} \times \text{Treated} \times \tilde{s}_{ie} + \phi_{ie} + \omega_t + \gamma_m + \varepsilon_{iet}. \quad (\text{A6})$$

Because the score is centered, δ_0^B is the treated-control post contrast at the mean predicted intention. The coefficient β^S is the common post-score gradient, while δ^S is the change in the treated-control post contrast associated with a one-unit increase in predicted purchase intention. Table A14 reports the complete continuous-score estimates for both outcomes.

Table A14: Complete continuous predicted-intention specifications

	Purchase frequency (1)	Novelty-seeking (2)
Post $\times$ Treated	.0023 (.0028) [.4337]	.0190 (.0151) [.2265]
Post $\times$ Centered predicted-intention score	-.0568 (.0046) [<.001]	.0533 (.0149) [.0024]
Post $\times$ Treated $\times$ Centered predicted-intention score	.0076 (.0051) [.1522]	-.0452 (.0276) [.1199]
Observations	321,184	99,644
R^2	.565	.416
Within R^2	.018	.000
Consumer fixed effects	Yes	Yes
Relative-period fixed effects	Yes	Yes
Calendar-month fixed effects	Yes	Yes

Notes: The predicted-intention score is centered at its estimation-sample mean. Post $\times$ Treated therefore gives the treated-control post contrast at the mean score. The score coefficient is the post-period score gradient common to treated and control consumers. The triple interaction, shown in boldface, is the focal coefficient and shows how the treated-control post contrast changes with the predicted-intention score. All remaining conventions and outcome definitions follow Table A13. Standard errors, clustered by closure event (18 clusters), are in parentheses; p-values are in brackets.

The continuous estimates in Table A14 preserve the binary results' sign pattern. At the mean score, the

treated-control post contrasts are 0.0023 for purchase frequency and 0.0190 for novelty-seeking, and neither is statistically distinguishable from zero. The common post-score gradients are -0.0568 and 0.0533 , respectively, indicating that higher-score consumers have lower post-period purchase frequency but higher post-period novelty-seeking after pooling treatment groups. The focal treated-score interactions are 0.0076 for purchase frequency and -0.0452 for novelty-seeking. Thus, for a 10-percentage-point increase in the predicted-intention score, the fitted DDD changes by $+0.08$ percentage points per day for purchase frequency and -0.45 points for novelty-seeking. Neither slope interaction is statistically distinguishable from zero at the 5% level under closure-clustered inference. The purchase-frequency pretrend caveat remains, while the novelty-seeking score-pretrend test does not reject equality.

D.2 Exploration of Recently Introduced Products

The main novelty-seeking outcome defines exploration relative to the consumer’s own purchase history. As a robustness check, we instead ask whether consumers select products that have appeared recently in the transaction data. Let F_j be product j ’s first observed sale date and let $\mathcal{W}_{et}$ and $\mathcal{W}_{e,t-1}$ be the current and immediately previous event windows. We define new-product-seeking as

$$Y_{iet}^{NP} = \frac{|\{j \in \mathcal{J}_{iet} : F_j \in \mathcal{W}_{e,t-1} \cup \mathcal{W}_{et}\}|}{|\mathcal{J}_{iet}|}. \quad (\text{A7})$$

This measure is the share of distinct products purchased in the current window whose first observed sale falls in the current or immediately previous event window. It proxies for recently introduced products, but first observed sale need not be the formal catalog launch date. Like novelty-seeking, it is defined only in windows with a purchase. Table A15 reports the binary group decomposition and the complete binary and continuous-score specifications.

Panel A of Table A15 yields a 0.80-point low-intention effect and a 3.37-point decline for high-predicted-intention consumers. Their difference is -4.17 percentage points (SE 0.89; $p < 0.001$), or 28.6% of the pre-period mean. Panel B shows that the binary specification also contains a positive common post shift of 3.00 percentage points, which is not a treatment effect. In the continuous specification, the treated-control contrast at the mean score is -0.33 points and is not statistically distinguishable from zero, while the common post-score gradient is 4.68 points ($p < 0.001$). The focal continuous-score interaction is -4.86 points, so a 10-percentage-point increase in predicted purchase intention lowers the treated-control post contrast by

Table A15: Robustness to using new-product-seeking

	Binary high-intention (1)	Continuous intention score (2)
<i>Panel A. Group effects and incremental interruption contrast</i>		
Low predicted purchase intention	.0080 (.0113) [.4871]	—
High predicted purchase intention	-.0337 (.0145) [.0331]	—
High minus low DDD	-.0417 (.0089) [<.001]	—
<i>Panel B. Complete regression coefficients</i>		
Post × Treated	.0080 (.0113) [.4871]	-.0033 (.0120) [.7851]
Post × Predicted intention	.0300 (.0045) [<.001]	.0468 (.0099) [<.001]
Post × Treated × Predicted intention	-.0417 (.0089) [<.001]	-.0486 (.0156) [.0063]
Pre-period outcome mean	.1456	.1456
Observations	99,644	99,644
R^2	.393	.393
Within R^2	.0008	.0007
Consumer fixed effects	Yes	Yes
Relative-period fixed effects	Yes	Yes
Calendar-month fixed effects	Yes	Yes

Notes: Column (1) uses the binary high-predicted-intention indicator; column (2) uses the mean-centered predicted-intention score. In Panel A, the low- and high-intention rows are directly estimated treated-control post contrasts, and the high-minus-low row is their DDD. These discrete group effects are not defined for column (2). In Panel B, Post × Predicted intention is the common post shift across treated and control consumers; it is not a treatment effect. Boldface marks the focal DDD coefficients. In column (2), a one-unit change corresponds to moving across the full predicted-intention scale. Standard errors, clustered by closure event (18 clusters), are in parentheses; p-values are in brackets. All regressions omit the closure window.

0.49 percentage points ($p = 0.006$). The high-minus-low pretrend tests do not reject for either specification ($p = 0.114$ for the binary measure and $p = 0.377$ for the continuous score). The robustness check therefore supports the main novelty-seeking result using a definition tied to recently appearing products rather than the consumer's own history.

D.3 Novelty-Seeking among Non-Coffee Products

This exercise examines whether the main product-choice pattern is confined to choices among coffee products. We classify products using the commodity-description fields in the transaction data. Each product line belongs to one mutually exclusive source category. The broad non-coffee definition combines non-coffee beverages, food and other non-coffee items; the consumables definition retains beverages and food; and the beverage-only definition retains non-coffee drinks. We identify products jointly by source category and exact description so that identical descriptions appearing in different categories remain distinct. Within each scope, novelty-seeking is the share of distinct qualifying products that the consumer had not purchased previously and is observed only in windows containing a qualifying purchase.

Table A16 reports the collapsed DDD estimates and the corresponding sample-entry diagnostics.

Table A16: Novelty-Seeking among Non-Coffee Products

	All non-coffee	Non-coffee consumables	Non-coffee beverages
<i>Panel A. Novelty-seeking conditional on purchase</i>			
High-minus-low DDD	−4.13 (3.40)	−4.11 (3.39)	−6.26 (4.12)
CRV1 <i>p</i> -value	.242	.242	.147
Restricted wild-cluster <i>p</i> -value	.225	.230	.119
Joint pretrend <i>p</i> -value	.006	.006	.226
Observations	30,444	30,418	21,433
<i>Panel B. Probability of any qualifying purchase</i>			
High-minus-low DDD	−1.70 (1.00)	−1.75 (1.01)	−2.24 (0.90)
CRV1 <i>p</i> -value	.107	.102	.023
Joint pretrend <i>p</i> -value	.093	.088	.121
Observations	321,184	321,184	321,184

Notes: Estimates and standard errors are in percentage points. All non-coffee combines non-coffee beverages, food and other non-coffee items. Non-coffee consumables combines beverages and food. Panel A defines novelty-seeking as the share of distinct products in the selected scope that the consumer had not purchased before the window; it is observed only in windows containing a qualifying purchase. Panel B uses the full consumer-period panel and indicates whether the window contains such a purchase. All regressions use the same binary DDD specification, fixed effects and 18 closure-event clusters as the main analysis. The joint pretrend row tests whether the DDD coefficients in relative periods −4, −3 and −2 are jointly zero, with period −1 omitted.

The collapsed DDDs are negative across all three product scopes but imprecisely estimated. The all-non-coffee estimate is −4.13 percentage points, with a 95% confidence interval from −11.31 to 3.05 points. Removing non-consumable items has essentially no effect because these products account for few qualifying purchasing windows. The beverage-only estimate is more negative, but its confidence interval also includes zero.

The dynamic and sample-entry diagnostics prevent a causal interpretation of these estimates. For the all-non-coffee and consumables outcomes, the joint lead tests reject parallel triple trends. The rejection is driven primarily by relative period -3 , in which the event-study DDD is $+10.33$ percentage points ($p = 0.005$); the post-reopening coefficients also alternate in sign. Panel B shows that the qualifying-purchase DDDs are -1.70 and -1.75 percentage points for the two broader scopes, but neither is statistically distinguishable from zero. The beverage-only novelty specification does not reject its pretrend, but the corresponding entry DDD is -2.24 percentage points ($p = 0.023$), showing that high-predicted-intention treated consumers become differentially less likely to make a qualifying purchase. Because novelty-seeking is conditional on such a purchase, this change may alter the composition of the observed sample. The exercise therefore provides a product-scope sensitivity check rather than independent confirmation of the main causal result.

D.4 Accounting for Local Competition

The store-address data report the number of potentially competing stores located within 500 and 1,500 meters of each of the chain’s 260 stores. We use the 500-meter count as our primary measure of local competition and the 1,500-meter count to assess sensitivity to the geographic radius. These measures are available for all 108 stores in the closure-event design. We assign the counts to each consumer through the consumer’s preferred store in the pre-closure period, reconstructed using the same five-purchase-day and 80% purchase-share requirements as in the main analysis. The data source does not identify the brands or operating status of nearby establishments, the underlying point-of-interest provider, or the date on which the counts were measured. The counts are therefore time-invariant proxies for the local competitive environment; they do not directly measure other-brand competition or temporary competitor closures during the study period.

The observed competitor counts differ between treated and control stores but have substantial overlapping support. Treated stores have an average of 6.00 potentially competing stores within 500 meters, compared with 12.54 among control stores, yielding a treated-minus-control standardized mean difference of -0.397 . The observed ranges overlap from 0 to 19 stores, an interval that contains all 18 treated stores and 75 of the 90 distinct control stores. To allow this time-invariant characteristic to affect the post-reopening comparison, let $\tilde{C}_s$ denote the standardized competitor count assigned through preferred store s . We add equation (13).

The original Post-by-Treated, Post-by-High, and triple-interaction terms remain in the regression. Be-

cause the competitor count is standardized across unique consumers, the coefficient on $\text{Post} \times \text{Treated} \times \text{High}$ is the fitted DDD at the mean observed level of local competition. At a standardized competitor count c , the fitted DDD is $\delta^D + \kappa_{THC}c$, and its standard error incorporates the covariance between the two coefficients. Table A17 reports the fitted DDDs under alternative measures and at prespecified points of the primary 500-meter count.

Table A17: Novelty-seeking DDDs across measures of local competition

Specification or competition level	Estimate	SE	95% CI	Wild-cluster p -value
<i>Panel A. Collapsed specifications</i>				
Paper-facing baseline	−4.15	1.84	[−8.04, −0.26]	.045
Raw 500m competitor count, consumer mean	−3.33	1.43	[−6.35, −0.31]	.090
$\log(1 + 500\text{m competitor count})$, consumer mean	−3.92	1.76	[−7.64, −0.20]	.065
Raw 1,500m competitor count, consumer mean	−4.04	1.64	[−7.51, −0.57]	.038
$\log(1 + 1,500\text{m competitor count})$, consumer mean	−4.11	1.59	[−7.46, −0.76]	.029
500m competitor count ≤ 3 (descriptive)	−5.39	3.65	[−13.17, 2.40]	.178
Preferred-store $\times$ event $\times$ post fixed effects	−4.45	1.71	[−8.07, −0.84]	.016
<i>Panel B. Fitted DDDs in the raw 500-meter specification</i>				
500m competitor count = 2	−5.03	3.15	[−11.67, 1.61]	.161
500m competitor count = 5	−4.39	2.42	[−9.50, 0.71]	.109
500m competitor count = 13	−2.70	1.25	[−5.35, −0.06]	.213
Change in DDD per SD of local competition	3.00	3.86	[−5.15, 11.14]	.490

Notes: Estimates and standard errors are in percentage points. The specifications that interact the DDD with local competition include the complete Post-interaction hierarchy in equation (13). Competitor counts of 2, 5, and 13 are the 25th percentile, median, and 75th percentile among the 108 design stores. All full-sample regressions contain 99,644 observations and 18 closure-event clusters. The low-competition specification contains 41,175 observations and 16 clusters. The preferred-store fixed-effect specification adds preferred-store-by-event-by-post fixed effects, which absorb additive local shocks common to high- and low-intention consumers assigned to the same store. Standard errors and confidence intervals use CRV1 clustering by closure event. Restricted wild-cluster p -values use 9,999 Rademacher draws.

Panel A of Table A17 shows that the alternative measures of local competition preserve a negative fitted DDD at their respective consumer means. Relative to the −4.15-point main estimate, the fitted DDD is −3.33 points using the raw 500-meter count and −3.92 points using its logarithm. The corresponding estimates are −4.04 and −4.11 points using the raw and logged 1,500-meter counts. Their restricted wild-cluster p -values are 0.090, 0.065, 0.038, and 0.029, respectively. Thus, the negative sign is consistent across specifications, but statistical significance under few-cluster inference is not. The low-competition restriction yields a more negative but imprecise estimate of −5.39 points ($p = 0.178$), while preferred-store-by-event-by-post fixed effects yield an estimate of −4.45 points ($p = 0.016$). Panel B evaluates the raw 500-meter specification at counts of 2, 5, and 13. The fitted DDD remains negative at −5.03, −4.39, and −2.70 points, respectively, but the corresponding wild-cluster p -values are 0.161, 0.109, and 0.213. A

one-standard-deviation increase in local competition changes the fitted DDD by an estimated +3.00 points ($p = 0.490$), and the 95% confidence interval ranges from -5.15 to 11.14 points. Thus, the estimated relationship between local competition and the DDD is imprecise.

Removing one closure and its five matched controls at a time produces fitted DDDs ranging from -7.10 to -3.18 percentage points at a count of 2 and from -3.25 to -1.15 percentage points at a count of 13. Thus, no single closure accounts for the negative sign. Replacing consumer-history novelty with new-product-seeking also preserves the pattern: the fitted DDD is -3.85 percentage points at the mean 500-meter competitor count and ranges from -4.12 to -3.10 percentage points across counts of 2, 5, and 13.

Two additional checks address overlap in the observed local competitive environment and unobserved additive local shocks. First, a descriptive restriction to stores with no more than three potentially competing stores retains 42 design stores, including 9 treated stores, and yields a DDD of -5.39 percentage points. This comparison has 16 contributing closure clusters, and only 8 matched sets contain qualifying stores on both treatment sides, so its confidence interval is necessarily wide. Second, preferred-store-by-event-by-post fixed effects absorb arbitrary additive post-reopening shocks shared by consumers assigned to the same preferred store. The resulting DDD is -4.45 percentage points. These fixed effects do not, however, absorb a local shock that affects high- and low-predicted-intention consumers differently within the same store.

The event-study diagnostics limit a causal interpretation of how local competition moderates the treatment effect. We interact the complete local-competition hierarchy with relative-period indicators, using period -1 as the reference. The estimated competition-interaction coefficients in periods -4 , -3 , and -2 are 0.1309 , 0.0906 , and 0.0695 , and a joint closure-clustered test rejects equality to zero ($p = 0.002$). By comparison, an event study with preferred-store-by-event-by-period fixed effects does not reject the three DDD leads jointly ($p = 0.880$). Taken together, these results show that the negative collapsed DDD is not confined to stores with high observed competitor counts, while the differential pre-closure movement prevents the auxiliary measure from identifying a distinct competition mechanism.